

Distance-Residual Physics-Informed Neural Networks: A Deep Learning Framework for Differential and Partial Differential Inclusions

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Abstract

We introduce *Distance-Residual Physics-Informed Neural Networks* (DR-PINNs), a physics-informed learning framework for approximating solutions of differential and partial differential inclusions, governing laws in which a differential operator is constrained to lie in a set-valued map rather than equaling a prescribed function. The method replaces the classical point-wise PDE/ODE residual by the squared distance from the differential operator to the admissible set. This distance vanishes exactly when the inclusion is satisfied and measures the infimal correction needed for the operator to enter the admissible set. For a fixed closed convex admissible set, the squared distance is differentiable with respect to the operator value, with gradient given by the metric projection. When the admissible set also depends on the network state, that dependence is included through the chain rule. The framework encompasses ordinary differential inclusions and partial differential inclusions with set-valued reaction terms. For both settings we prove *consistency*: under the stated closedness, measurability, convexity, and growth assumptions, any sequence of candidates satisfying the initial (and, in the parabolic case, boundary) conditions whose continuous distance-residual

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functional tends to zero admits a subsequence converging to an exact solution of the target inclusion. These are conditional statements for the continuous distance-residual functional; they do not cover the finite-collocation training loss, the behaviour of the optimizer, or convergence rates. The proofs combine a priori estimates (Grönwall inequality and parabolic energy estimates), weak compactness arguments (Arzelà–Ascoli and an Aubin–Lions-type compact embedding), and the lower semicontinuity of normal convex integrands. For admissible sets given as convex hulls of finitely many vertices, projection onto the set reduces to a small convex quadratic program, making the loss efficiently computable inside the training loop. Numerical experiments demonstrate high accuracy on the differential-inclusion benchmarks; in the parabolic relay benchmark, trained against an ε -banded relaxation of the relay, the learned solutions reproduce the finite-time extinction dynamics quantitatively in the band-entry observable (seed-stable, persistent, and matching a reference solver to the resolution of the evaluation grid) while the distance residual remains concentrated in a thin layer ahead of the discontinuous extinction front.

Keywords: Physics-Informed Neural Networks, distance function, differential inclusions, partial differential inclusions, consistency analysis.
2020 MSC: 34A60, 49J53, 65M99, 68T07.

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1. Introduction

Physics-Informed Neural Networks (PINNs) approximate solutions of differential equations by incorporating the governing law directly into the training loss [23]. Since the work of Raissi, Perdikaris, and Karniadakis, they have been used for nonlinear PDEs in fluid mechanics, heat transfer, and wave propagation, as well as inverse problems, parameter identification, operator learning, and high-dimensional stochastic differential equations. In these settings the governing law is single-valued, for example an ODE $\dot{x}(t) = f(t, x(t))$ or a PDE $\mathcal{D}(u) = f$, and the PINN loss is built from the pointwise residual obtained after substituting a neural network surrogate into that equality.

The same residual construction is not available when the governing law is set-valued:

$$\mathcal{D}(u) \in \mathcal{F}(u),$$

where $\mathcal{F}(u)$ is a multifunction (set-valued map). Concrete instances include:

- *Differential inclusions*: $\dot{x}(t) \in F(t, x(t))$, where $F(t, x)$ is a set of admissible velocities;
- *Parabolic partial differential inclusions*: $\partial_t u - \Delta u \in \Phi(u)$, where $\Phi(u)$ is a set-valued reaction term.

In these cases there is no single-valued right-hand side against which to form the usual residual. For a set-valued law $\mathcal{D}(u) \in \mathcal{F}(u)$, consider instead the quantity

$$\text{dist}(\mathcal{D}(u), \mathcal{F}(u)) := \inf_{v \in \mathcal{F}(u)} \|\mathcal{D}(u) - v\|$$

which is well defined whenever $\mathcal{F}(u)$ is nonempty and closed, and it vanishes if and only if the inclusion is satisfied. This suggests replacing the classical PINN residual by a *distance residual* and training a neural network to minimize the loss

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \text{dist}^2(\mathcal{D}(u_\theta)(z_i), \mathcal{F}(u_\theta)(z_i)),$$

where u_θ is a neural network surrogate of u , and $\{z_i\}_{i=1}^N$ are collocation points. The resulting methodology, which we call *Distance-Residual Physics-Informed Neural Networks* (DR-PINNs), extends the PINN paradigm to genuinely set-valued differential laws, while reducing to the classical formulation when $\mathcal{F}(u) = \{f(u)\}$ is a singleton.

Related work. Distance and projection operators have also been used in PINNs, but for different purposes. Deguchi and Asai [10] use R -function-based approximate distance fields to impose Dirichlet boundary conditions exactly on non-convex domains, so the distance encodes the geometry of the boundary rather than an admissible set for the dynamics. Several recent architectures equip a PINN with a projection layer that maps an unconstrained network output onto a feasible region to guarantee hard-constraint satisfaction; see the KKT-based projection of Iftakher et al. [15] or the integral-conservation projection of Baez et al. [4]. In all of these constructions the projection acts once, on the network output, to enforce a fixed constraint; it does not serve as the training residual, and none of them addresses a genuinely set-valued evolution law.

Wu, Lisser, and Chamoin [24, 25, 26, 27] study variational inequalities and complementarity problems using neurodynamic optimization and PINNs. They first rewrite the variational inequality as an ODE whose right-hand side contains a metric projection onto the fixed feasible set, and then train a PINN with the ordinary pointwise residual of that auxiliary equation. In this formulation the set-valued aspect is removed before training. DR-PINNs instead leave $\dot{x}(t) \in F(t, x(t))$ multivalued and measure the distance to the state-dependent set $F(t, x_\theta(t))$, so the projection is recomputed along the current trajectory during training.

Kasyanov, Kapustyan, Levenchuk, and Novykov [18] proposed a machine-learning method (including PINNs and the deep Galerkin method) for reaction-diffusion equations with *multivalued* interaction functions. Their method first replaces the multifunction by a single-valued Lipschitz approximation (a Pasch–Hausdorff-type regularization), and a classical pointwise residual is then trained for the regularized, single-valued equation; the set-valued character of the law is thus removed *before* training, and the quality of the approximation is governed by an additional regularization parameter. A follow-up study [19] analyzes the Monte Carlo estimation of the corresponding loss functionals and shows that the variance of the standard estimator degenerates as the regularization parameter grows, which further underlines the cost of the prior single-valued regularization. DR-PINNs instead keep the inclusion genuinely multivalued and use the distance to the (possibly state-dependent) admissible set itself as the training residual, with no regularization parameter to drive to a limit; this requires the evaluation of a metric projection inside the training loop. A direct experimental comparison with the regularization-based approach of [18] on common benchmarks remains to be carried out.

The theory of differential inclusions is well developed [2, 3, 11], whereas

PINN-type methods for set-valued problems have so far, to the best of our knowledge, used a prior single-valued regularization of the multifunction [18, 19]. The contribution here is to use the distance from the differential operator to the admissible set directly as the training residual, without prior single-valued regularization, and to prove consistency results for both ordinary and parabolic differential inclusions.

Contributions. The paper makes the following contributions:

- (a) A general *distance-residual principle* (Section 3) that provides a unified method for constructing PINN losses for set-valued differential laws with nonempty closed values. The later properties use stronger hypotheses: differentiability of the loss and computability of the projection require closed *convex* values, while the consistency results additionally use the measurability, growth, and compactness assumptions stated in Sections 4.3 and 5.
- (b) *Conditional consistency theorems* for both ordinary differential inclusions (Section 4.3) and partial differential inclusions (Section 5.2), showing that vanishing *continuous* distance-residual functionals certify admissible solutions. Their relation to finite-collocation training is discussed in Remark 3.2; the theorems do not address optimizer convergence or convergence rates.
- (c) A *practical projection algorithm* (Section 4.2) for admissible sets given as convex hulls, based on a small quadratic program combined with the Quickhull algorithm.
- (d) *Numerical experiments* (Section 6) on benchmark differential and partial differential inclusions.

2. Preliminaries

2.1. Set-valued functions

We collect the notation for set-valued functions used below, including the notions of graph and semicontinuity.

Definition 2.1. Let X and Y be nonempty sets. A *set-valued function* (or *multifunction*) from X to Y is a mapping

$$\mathcal{F} : X \rightrightarrows Y,$$

which assigns to each $x \in X$ a subset $\mathcal{F}(x) \subseteq Y$. The *graph* of $\mathcal{F}$ is defined as

$$\text{gr}(\mathcal{F}) := \{(x, y) \in X \times Y : y \in \mathcal{F}(x)\}.$$

For a subset $A \subseteq X$, the image of A under $\mathcal{F}$ is given by

$$\mathcal{F}(A) := \bigcup_{x \in A} \mathcal{F}(x).$$

Definition 2.2. Let X and Y be topological spaces, and $\mathcal{F} : X \rightrightarrows Y$ a set-valued map. We say that $\mathcal{F}$ is *upper semicontinuous* at $x \in X$ whenever for every open set U containing $\mathcal{F}(x)$ there exists a neighborhood V of x such that

$$\mathcal{F}(V) \subset U.$$

The map $\mathcal{F}$ is said to be *lower semicontinuous* at $x \in X$ if for every $y \in \mathcal{F}(x)$ and every neighborhood U of y , there exists a neighborhood V of x such that

$$\mathcal{F}(x') \cap U \neq \emptyset \quad \text{for every } x' \in V.$$

$\mathcal{F}$ is said to be upper semicontinuous (respectively, lower semicontinuous) if it is upper semicontinuous (respectively, lower semicontinuous) at every point $x \in X$. Finally, $\mathcal{F}$ is called *continuous* if it is both upper and lower semicontinuous.

Following [3], we formulate properties of a set-valued map in terms of its graph: the map is said to satisfy property P when its graph has that property. For instance, a set-valued map is said to be closed (respectively convex, measurable) if and only if its graph is closed (respectively convex, measurable).

In contrast, $\mathcal{F}$ is said to be *closed-valued*, *convex-valued*, *bounded-valued*, *compact-valued*, or *strict* if for every $x \in X$, the set $\mathcal{F}(x)$ is closed, convex, bounded, compact, or nonempty, respectively. It is easy to show that every strict, closed, set-valued function $\mathcal{F} : X \rightrightarrows Y$ between metric spaces is closed-valued (see, e.g., [?, p. 22]).

We conclude by recalling the notions of selection and integral of a set-valued function.

Definition 2.3. Let X and Y be nonempty sets and assume that $\mathcal{F} : X \rightrightarrows Y$ is a strict set-valued function. A selection from $\mathcal{F}$ is a function $f : X \rightarrow Y$ such that $f(x) \in \mathcal{F}(x)$ for every $x \in X$.

Definition 2.4. Let $I \subseteq \mathbb{R}$ be an interval and assume that $\mathcal{F} : I \rightrightarrows \mathbb{R}^d$ is strict and upper semicontinuous. The integral of $\mathcal{F}$ over I is defined as the set

$$\int_I \mathcal{F}(s) ds := \left\{ \int_I f(s) ds : f \text{ is an integrable selection from } \mathcal{F} \right\}.$$

2.2. Projection onto a set

Fix a normed space X . If $C \subseteq X$ is a nonempty closed set, then the infimum

$$\text{dist}(x_0, C) := \inf\{\|x_0 - x\|_X : x \in C\}, \quad x_0 \in X,$$

is always well defined; it is attained, for instance, whenever X is finite-dimensional or, more generally, whenever C is boundedly compact (in a general infinite-dimensional normed space a nearest point may fail to exist). We say that $z \in C$ is a *projection* of x_0 onto C whenever $\text{dist}(x_0, C) = \|z - x_0\|_X$. In general there can be several projections of x_0 onto C , that is, there might be more than one point in C closest to x_0 . A fundamental result in convex analysis (see, e.g., [9, Theorem 5.2, p. 132]) establishes that, whenever X is a Hilbert space, the projection is unique provided that C is nonempty, closed, and convex.

In the case $X = \mathbb{R}^n$, we assume throughout this work that the norm is the Euclidean norm $\|\cdot\|_2$. To simplify the notation, we write $|\cdot|$ instead of $\|\cdot\|_2$. In this scenario, if $C \subset \mathbb{R}^n$ is nonempty, closed, and *convex*, then the Euclidean projection of x_0 onto C is attained and unique, and will be denoted by $\Pi_C(x_0)$. In other words,

$$\Pi_C(x_0) := \operatorname{argmin}_{x \in C} |x_0 - x|. \quad (2.1)$$

For a nonempty closed set that is *not* convex, a nearest point still exists in $\mathbb{R}^n$, but it need not be unique; whenever we write Π_C below, the set C is assumed to be convex.

2.3. Background on Differential Inclusions

Differential inclusions extend ordinary differential equations by allowing the velocity of the state variable to belong to a set of admissible directions. Let $T > 0$, $x_0 \in \mathbb{R}^d$, and assume that

$$F : [0, T] \times \mathbb{R}^d \rightrightarrows \mathbb{R}^d$$

is a set-valued map. Consider the initial value problem (IVP) for a *differential inclusion*:

$$\dot{x}(t) \in F(t, x(t)), \quad x(0) = x_0. \quad (2.2)$$

For each pair (t, x) , the set $F(t, x)$ represents the admissible velocities available to the system at time t and state x .

Definition 2.5. A function $x : [0, T] \rightarrow \mathbb{R}^d$ is a *solution* of the IVP (2.2) if it is absolutely continuous, satisfies $x(0) = x_0$, and $\dot{x}(t) \in F(t, x(t))$ for almost every $t \in [0, T]$.

Absolute continuity guarantees that $\dot{x}(t)$ exists almost everywhere. Thus, for any solution x , we have the exact representation

$$x(t) = x_0 + \int_0^t \dot{x}(s) ds.$$

However, to translate the inclusion $\dot{x}(t) \in F(t, x(t))$ into an integral characterization in terms of F , regularity conditions are needed (see Lemma 2.6 below).

Lemma 2.6. (Integral Representation, [2, Lemma 1, p.99]) *Assume that $F : [0, T] \times \mathbb{R}^d \rightrightarrows \mathbb{R}^d$ is upper semicontinuous, compact-valued, and convex-valued. Then, a continuous function $x(\cdot)$ is a solution on $[0, T]$ of the differential inclusion*

$$\dot{x}(t) \in F(t, x(t))$$

if and only if for every pair $(t_1, t_2) \in [0, T]$ with $t_1 < t_2$,

$$x(t_2) \in x(t_1) + \int_{t_1}^{t_2} F(s, x(s)) ds.$$

The proof of the following result appears in [11, Theorem 5.2, p.58].

Theorem 2.7 (Basic existence). *Let $F : [0, T] \times \mathbb{R}^d \rightrightarrows \mathbb{R}^d$ be strict, closed-valued, and convex-valued. Assume that $F(t, x)$ is measurable in t for each $x \in \mathbb{R}^d$, upper semicontinuous in x for each $t \in [0, T]$, and satisfies the linear growth condition*

$$\sup_{v \in F(t, x)} |v| \leq m(t)(1 + |x|) \quad (2.3)$$

for some $m \in L^1(0, T)$ and each $(t, x) \in [0, T] \times \mathbb{R}^d$. Then, for every $x_0 \in \mathbb{R}^d$, there exists a solution of (2.2).

Remark 2.8. Strictness, closedness, and convexity of F support the measurable-selection and fixed-point arguments used in the existence theory; see [2, 3, 11]. They also ensure the properties of the distance to $F(t, x)$ used later: finiteness, attainment, and, under convexity, differentiability.

Unlike ordinary differential equations, uniqueness of solutions of (2.2) is not expected in general. For a given initial condition x_0 , there may exist many admissible trajectories. The *solution set*

$$\mathcal{S}(x_0) = \{x \in \text{AC}([0, T]; \mathbb{R}^d) : x(0) = x_0, \dot{x}(t) \in F(t, x(t)) \text{ a.e. on } [0, T]\}, \quad (2.4)$$

where $\text{AC}([0, T]; \mathbb{R}^d)$ denotes the collection of absolutely continuous functions from $[0, T]$ into $\mathbb{R}^d$, is naturally set-valued. Consequently, a trained DR-PINN approximates *one* admissible trajectory, not a distinguished one; this interpretation is made precise by the consistency result in Section 4.3.

2.4. Background on Parabolic Partial Differential Inclusions

Next, we consider partial differential inclusions. Although, for the sake of simplicity, in what follows we only address the parabolic case, most of the main ideas and results extend to other types of partial differential inclusions.

Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with $C^{1,1}$ boundary, or a bounded convex domain, let $T > 0$, and set $Q := (0, T) \times \Omega$. Consider the initial-boundary value problem (IBVP) for the parabolic partial differential inclusion:

$$\begin{cases} \partial_t u(t, x) - \Delta u(t, x) \in \Phi(u(t, x)), & (t, x) \in Q, \\ u(0, x) = u_0(x), & x \in \Omega, \\ u(t, x) = 0, & (t, x) \in (0, T) \times \partial\Omega, \end{cases} \quad (2.5)$$

where $\Phi : \mathbb{R} \rightrightarrows \mathbb{R}$ is a set-valued reaction term and $u_0 \in H_0^1(\Omega)$.

The natural functional framework for strong solutions of (2.5) is the *parabolic energy space*

$$\mathcal{W}(0, T) := L^2(0, T; H^2(\Omega) \cap H_0^1(\Omega)) \cap H^1(0, T; L^2(\Omega)),$$

equipped with the norm

$$\|u\|_{\mathcal{W}}^2 := \|u\|_{L^2(0, T; H^2(\Omega))}^2 + \|\partial_t u\|_{L^2(Q)}^2.$$

Setting $V := H^2(\Omega) \cap H_0^1(\Omega)$ and $W := L^2(\Omega)$, both Hilbert spaces, we have

$$\mathcal{W}(0, T) = \left\{ v \in L^2(0, T; V) : \frac{dv}{dt} \in L^2(0, T; W) \right\},$$

so by [7, Theorem II.5.14], the embedding

$$\mathcal{W}(0, T) \hookrightarrow C^0([0, T]; [V, W]_{1/2})$$

is continuous, where $[V, W]_{1/2}$ denotes the interpolation space of order $1/2$ between V and W (see also the intermediate-derivatives and trace theory of [21, Chapter 1, Theorem 3.1]). Under our standing regularity assumptions on Ω , the interpolation space is identified by the following lemma, which yields the continuous embedding

$$\mathcal{W}(0, T) \hookrightarrow C([0, T]; H_0^1(\Omega)). \quad (2.6)$$

Lemma 2.9 (Interpolation identity). *Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with $C^{1,1}$ boundary, or a bounded convex domain. Then,*

$$[H^2(\Omega) \cap H_0^1(\Omega), L^2(\Omega)]_{1/2} = H_0^1(\Omega). \quad (2.7)$$

Proof. Apply the abstract framework of [21, Chapter 1, §2.1] with $X = H_0^1(\Omega)$ (equipped with the gradient inner product) and $Y = L^2(\Omega)$. The associated positive self-adjoint operator ([21, Chapter 1, §2.1, eq. (2.4)]) is precisely the Dirichlet Laplacian $-\Delta_D$, i.e., the negative Laplacian $-\Delta$ restricted to the domain $D(-\Delta_D) = \{u \in H_0^1(\Omega) : -\Delta u \in L^2(\Omega) \text{ weakly}\}$. By [21, Chapter 1, §2.4, Proposition 2.1, eq. (2.42)],

$$[D(-\Delta_D), L^2(\Omega)]_{1/2} = H_0^1(\Omega).$$

It therefore remains to identify $D(-\Delta_D) = H^2(\Omega) \cap H_0^1(\Omega)$, which is exactly the statement of elliptic H^2 -regularity for the Dirichlet Laplacian. This equality holds in both admissible regimes for Ω : if Ω is bounded and convex (with no smoothness assumed on the boundary), by [14, Theorem 3.2.1.2], and if Ω is bounded with $C^{1,1}$ boundary (not necessarily convex), by [14, Theorem 2.4.2.5], in both cases applied to $-\Delta$ with homogeneous Dirichlet conditions. Combining the two results yields (2.7). $\square$

The compactness statement needed below is stronger than the usual Aubin–Lions conclusion, which gives compactness in $L^2(0, T; L^2(\Omega))$. We state the required version separately.

Lemma 2.10 (Compact embedding into $C([0, T]; L^2(\Omega))$). *The embedding*

$$\mathcal{W}(0, T) \hookrightarrow C([0, T]; L^2(\Omega))$$

is compact.

Proof. Let $\{u_n\}_{n \in \mathbb{N}} \subset \mathcal{W}(0, T)$ be bounded, that is, there exists a constant $M > 0$ such that $\|u_n\|_{\mathcal{W}} \leq M$ for every $n \in \mathbb{N}$. First, for all $0 \leq s \leq t \leq$

T , the fundamental theorem of calculus in $L^2(\Omega)$ and the Cauchy–Schwarz inequality give

$$\begin{aligned}\|u_n(t) - u_n(s)\|_{L^2(\Omega)} &\leq \int_s^t \|\partial_t u_n(\tau)\|_{L^2(\Omega)} d\tau \\ &\leq |t - s|^{1/2} \|\partial_t u_n\|_{L^2(Q)} \leq M |t - s|^{1/2},\end{aligned}$$

so the sequence is uniformly equicontinuous with values in $L^2(\Omega)$. Second, by the continuous embedding (2.6) there is a constant $C_e > 0$ such that $\sup_{t \in [0, T]} \|u_n(t)\|_{H_0^1(\Omega)} \leq C_e M$ for every n ; hence, for each fixed $t \in [0, T]$, the set $\{u_n(t) : n \in \mathbb{N}\}$ is bounded in $H_0^1(\Omega)$ and therefore, by the Rellich–Kondrachov theorem, relatively compact in $L^2(\Omega)$. The Arzelà–Ascoli theorem in the Banach space $C([0, T]; L^2(\Omega))$ then provides a uniformly convergent subsequence, which proves the claimed compactness. $\square$

The (compact, hence continuous) embedding into $C([0, T]; L^2(\Omega))$ ensures in particular that the initial condition $u(0, \cdot) = u_0$ is well defined pointwise in time for every $u \in \mathcal{W}(0, T)$.

Definition 2.11. A function $u \in \mathcal{W}(0, T)$ is a *strong solution* of (2.5) if

- (a) $u(0, \cdot) = u_0$ on Ω ,
- (b) $u(t, \cdot)|_{\partial\Omega} = 0$ for a.e. $t \in (0, T)$,
- (c) $\partial_t u(t, x) - \Delta u(t, x) \in \Phi(u(t, x))$ for a.e. $(t, x) \in Q$.

Since $u \in \mathcal{W}(0, T)$ implies $\partial_t u, \Delta u \in L^2(Q)$, the inclusion in (c) is meaningful pointwise almost everywhere. Moreover, by the continuous embedding (2.6), every $u \in \mathcal{W}(0, T)$ satisfies $u(t, \cdot) \in H_0^1(\Omega)$ for every $t \in [0, T]$, so condition (b) holds automatically (for every t , in fact) and is retained only for emphasis.

The *solution set* of (2.5) is

$$\mathcal{S}(u_0) := \{u \in \mathcal{W}(0, T) : u \text{ is a strong solution of (2.5)}\}. \quad (2.8)$$

As in the ODE setting, strong solutions need not be unique, i.e. $\mathcal{S}(u_0)$ may contain multiple elements.

The following existence result guarantees that $\mathcal{S}(u_0)$ is nonempty under standing assumptions on the set-valued reaction term $\Phi : \mathbb{R} \rightrightarrows \mathbb{R}$ that are *exactly* the hypotheses of the consistency analysis of Section 5.2:

- (P1) Φ is strict, closed-valued, and convex-valued.
- (P2) Φ is upper semicontinuous.
- (P3) Φ satisfies the linear growth condition (2.3) with $m \equiv m_0$ constant, i.e., there exists a constant $m_0 \geq 0$ such that

$$\sup_{r \in \Phi(s)} |r| \leq m_0(1 + |s|) \quad \forall s \in \mathbb{R}.$$

Since (P2) requires only *upper* semicontinuity, Φ need not admit any continuous selection; the relay term of Section 6.3 is a case in point. Selection theorems of Michael type (see, e.g., [11, Lemma 2.1]), which require lower semicontinuity, therefore cannot be used to reduce (2.5) to a single-valued semilinear equation. We instead work directly with the multivalued problem and obtain existence from the fixed-point theorem of Bohnenblust and Karlin [6], an extension of Kakutani's theorem [17] from compact convex sets in finite-dimensional spaces to closed convex sets in Banach spaces.

Linear solvability. For $h \in L^2(Q)$ and $u_0 \in H_0^1(\Omega)$, let $S(h) \in \mathcal{W}(0, T)$ denote the unique solution of the linear Dirichlet problem

$$\partial_t u - \Delta u = h \quad \text{on } Q, \quad u = u_0 \quad \text{on } \{0\} \times \Omega, \quad u = 0 \quad \text{on } (0, T) \times \partial\Omega.$$

By parabolic maximal L^2 -regularity, S is well defined and there exists a constant $C_{\text{par}} > 0$ such that

$$\|S(h)\|_{\mathcal{W}(0, T)} \leq C_{\text{par}}(\|h\|_{L^2(Q)} + \|u_0\|_{H_0^1(\Omega)}); \quad (2.9)$$

see [12, Section 7.1.3, Theorem 5] for smooth domains; the H^2 elliptic regularity required under our standing assumptions on Ω is supplied by [14, Theorem 3.2.1.2] in the convex case and by [14, Theorem 2.4.2.5] in the $C^{1,1}$ -boundary case (cf. the proof of Lemma 2.9). We write S_0 for the solution operator of the same problem with zero initial datum and $w_0 := S(0) \in \mathcal{W}(0, T)$ for the solution of the zero-source problem with initial datum u_0 . Then, we have $S(h) = S_0(h) + w_0$; by (2.9) (applied with $u_0 = 0$), $S_0 : L^2(Q) \rightarrow \mathcal{W}(0, T)$ is a bounded linear operator, and, after composition with the continuous embedding $\mathcal{W}(0, T) \hookrightarrow L^2(Q)$, it is continuous from $L^2(Q)$ into $L^2(Q)$.

We begin by showing that, for any fixed $w \in L^2(Q)$, the Nemytskii-type multimap $(t, x) \mapsto \Phi(w(t, x))$ admits at least one $L^2(Q)$ -selection.

Proposition 2.12 (Nonemptiness of the selection set). *Assume (P1)–(P3) and let $w \in L^2(Q)$. Then*

$$S_{\Phi,w}^2 := \{v \in L^2(Q) : v(t,x) \in \Phi(w(t,x)) \text{ a.e. on } Q\}$$

is nonempty, closed, and convex in $L^2(Q)$.

Proof. Let $C \subset \mathbb{R}$ be closed. By (P2), the set $\{s \in \mathbb{R} : \Phi(s) \subset \mathbb{R} \setminus C\}$ is open, hence its complement

$$\Phi^{-1}(C) := \{s \in \mathbb{R} : \Phi(s) \cap C \neq \emptyset\}$$

is closed, and in particular Borel. Writing an open set $U \subset \mathbb{R}$ as a countable union of closed sets shows that $\Phi^{-1}(U)$ is Borel for every open U ; thus Φ is a Borel measurable multifunction with nonempty closed values, by (P1).

For every closed $C \subset \mathbb{R}$,

$$\{(t,x) \in Q : \Phi(w(t,x)) \cap C \neq \emptyset\} = w^{-1}(\Phi^{-1}(C))$$

is Lebesgue measurable, being the preimage of a closed (hence Borel) set under the Lebesgue measurable function $w : Q \rightarrow \mathbb{R}$. Hence $F(t,x) := \Phi(w(t,x))$ is a measurable multifunction from the complete σ -finite measure space $(Q, \mathcal{L}(Q))$ into the nonempty closed subsets of $\mathbb{R}$. Here and elsewhere throughout this work, $\mathcal{L}(Q)$ denotes the collection of all Lebesgue measurable subsets of Q .

By the measurable selection theorem [3, Theorem 8.1.3], there exists a measurable $v : Q \rightarrow \mathbb{R}$ with $v(t,x) \in \Phi(w(t,x))$ a.e. on Q . By (P3), $|v(t,x)| \leq m_0(1 + |w(t,x)|)$ a.e., hence $|v|^2 \leq 2m_0^2(1 + |w|^2)$ a.e. and, since $|Q| < \infty$,

$$\|v\|_{L^2(Q)}^2 \leq 2m_0^2(|Q| + \|w\|_{L^2(Q)}^2) < \infty. \quad (2.10)$$

Therefore $v \in S_{\Phi,w}^2$, so $S_{\Phi,w}^2 \neq \emptyset$.

Convexity of $S_{\Phi,w}^2$ follows from the pointwise convexity of $\Phi(w(t,x))$ in (P1). For closedness, if $v_n \in S_{\Phi,w}^2$ and $v_n \rightarrow v$ in $L^2(Q)$, a subsequence converges a.e., and the pointwise closedness of $\Phi(w(t,x))$ gives $v \in S_{\Phi,w}^2$. $\square$

The fixed-point theorem we use is the following.

Theorem 2.13 (Bohnenblust–Karlin). *Let X be a Banach space and let $D \subset X$ be a nonempty closed convex set. Let $\mathcal{G} : D \rightrightarrows X$ be a strict, closed-valued, convex-valued multifunction such that $\mathcal{G}(D) \subset D$. Suppose that:*

- (a) *The graph of $\mathcal{G}$ is sequentially closed in $D \times X$: if $x_n \rightarrow x$ in D , $y_n \rightarrow y$ in X , and $y_n \in \mathcal{G}(x_n)$ for every n , then $y \in \mathcal{G}(x)$.*

(b) $\mathcal{G}(D)$ is contained in a sequentially compact subset of X .

Then $\mathcal{G}$ has a fixed point, i.e., there exists $x_0 \in D$ such that $x_0 \in \mathcal{G}(x_0)$.

Proof. This is [6, Theorem 4, p. 159], restated in the notation of the present paper. Bohnenblust and Karlin's proof uses the convexity of the values $\mathcal{G}(x)$ through the fact that the closed ε -neighbourhood of a convex set is again convex, which is what allows the finite-dimensional Kakutani theorem to be invoked on a finite ε -net, even though convexity of the values is not restated as a separately numbered hypothesis in their theorem statement; it is made explicit as a standing hypothesis in the modern formulation of [? , Corollary 9.8, p. 452], which we have followed above. $\square$

The key analytic ingredient, beyond Proposition 2.12, is the following weak-strong closedness property of the selection sets. When passing to the limit along $w_n \rightarrow w$ strongly in $L^2(Q)$ and $v_n \in S_{\Phi, w_n}^2$ such that $v_n \rightharpoonup v$ only weakly in $L^2(Q)$, it guarantees that the weak limit v remains an admissible selection of $\Phi(w(\cdot, \cdot))$; without it, one could not pass to the limit in the set-valued reaction term.

Lemma 2.14 (Weak-strong closedness of the selection sets). *Assume (P1)–(P3). Suppose that $\{w_n\}_{n \in \mathbb{N}}, \{v_n\}_{n \in \mathbb{N}} \subset L^2(Q)$ satisfy:*

- (i) $w_n \rightarrow w$ strongly in $L^2(Q)$,
- (ii) $v_n \in S_{\Phi, w_n}^2$ for every $n \in \mathbb{N}$,
- (iii) $v_n \rightharpoonup v$ weakly in $L^2(Q)$.

Then, $v \in S_{\Phi, w}^2$.

Proof. Since $w_n \rightarrow w$ strongly in $L^2(Q)$, we may pass to a subsequence, for which we use the same labeling as for the original sequence, along which $w_n(t, x) \rightarrow w(t, x)$ for a.e. $(t, x) \in Q$. Dropping finitely many terms does not affect a weak limit, so for every n the tail sequence $\{v_k\}_{k \geq n}$ also converges weakly to v . Applying Mazur's lemma (see, e.g., [9, Corollary 3.8]) to this tail sequence yields, for each n , a finite convex combination supported on indices $k \geq n$,

$$\hat{v}_n := \sum_{k=n}^{N(n)} \lambda_k^{(n)} v_k, \quad \lambda_k^{(n)} \geq 0, \quad \sum_{k=n}^{N(n)} \lambda_k^{(n)} = 1,$$

with $\|\hat{v}_n - v\|_{L^2(Q)} \leq 1/n$, so that $\hat{v}_n \rightarrow v$ strongly in $L^2(Q)$; passing to a further subsequence, not relabeled, we may also assume $\hat{v}_n(t, x) \rightarrow v(t, x)$ for a.e. $(t, x) \in Q$.

For each k , hypothesis (ii) gives $v_k(t, x) \in \Phi(w_k(t, x))$ for all (t, x) outside a null set $N_k \subset Q$; set $N := \bigcup_k N_k$, again a null set. Fix any $(t, x) \in Q \setminus N$ at which, in addition, both of the above a.e. convergences hold; such points form a set of full measure. Fix $\varepsilon > 0$. By (P1), $\Phi(w(t, x))$ is closed and convex, so $\Phi(w(t, x)) + \varepsilon B$, with B the open unit ball of $\mathbb{R}$, is open and convex. By (P2), Φ is upper semicontinuous at $w(t, x)$, so there exists $\delta > 0$ such that $|s - w(t, x)| < \delta$ implies $\Phi(s) \subset \Phi(w(t, x)) + \varepsilon B$. Since $w_k(t, x) \rightarrow w(t, x)$, there is $N_1 = N_1(t, x, \varepsilon)$ with

$$v_k(t, x) \in \Phi(w_k(t, x)) \subset \Phi(w(t, x)) + \varepsilon B, \quad k \geq N_1.$$

As $\Phi(w(t, x)) + \varepsilon B$ is convex, it contains every convex combination of the points $v_k(t, x)$ with $k \geq N_1$; since $\hat{v}_n$ is supported on indices $k \geq n$, we get $\hat{v}_n(t, x) \in \Phi(w(t, x)) + \varepsilon B$ for all $n \geq N_1$. Letting $n \rightarrow \infty$ and using $\hat{v}_n(t, x) \rightarrow v(t, x)$ together with the closedness of $\Phi(w(t, x)) + \varepsilon \overline{B}$, we obtain $v(t, x) \in \Phi(w(t, x)) + \varepsilon \overline{B}$. Since $\varepsilon > 0$ was arbitrary and $\Phi(w(t, x))$ is closed,

$$v(t, x) \in \bigcap_{\varepsilon > 0} (\Phi(w(t, x)) + \varepsilon \overline{B}) = \Phi(w(t, x)).$$

This holds for a.e. $(t, x) \in Q$, the exceptional set being a countable union of null sets, so $v \in S_{\Phi, w}^2$. $\square$

The solution multimap and an invariant set. We seek a strong solution as a fixed point of the set-valued map

$$\mathcal{G} : L^2(Q) \rightrightarrows L^2(Q), \quad \mathcal{G}(w) := \{ S_0(v) + w_0 : v \in S_{\Phi, w}^2 \}. \quad (2.11)$$

Indeed, if $w^* \in \mathcal{G}(w^*)$, then $w^* = S_0(v^*) + w_0$ for some $v^* \in S_{\Phi, w^*}^2$, and $u := w^* \in \mathcal{W}(0, T)$ solves $\partial_t u - \Delta u = v^* \in \Phi(w^*) = \Phi(u)$ a.e. on Q with $u(0) = u_0$, i.e. $u \in \mathcal{S}(u_0)$.

As is standard when the reaction term has linear (not small) growth, a self-mapping estimate for $\mathcal{G}$ in the ordinary $L^2(Q)$ -norm need not close, since the growth constant of (P3) enters the linear energy estimate with a factor that need not be smaller than 1. We circumvent this with a Bielecki-type exponentially weighted norm.

Lemma 2.15 (Weighted a priori bound). *Let $\lambda > 0$ and for every $g \in L^2(Q)$, define*

$$\|g\|_\lambda^2 := \int_0^T e^{-2\lambda t} \|g(t, \cdot)\|_{L^2(\Omega)}^2 dt.$$

Note that, since $e^{-2\lambda T} \leq e^{-2\lambda t} \leq 1$ on $[0, T]$, $\|\cdot\|_\lambda$ is a norm on $L^2(Q)$ equivalent to $\|\cdot\|_{L^2(Q)}$ with

$$e^{-\lambda T} \|g\|_{L^2(Q)} \leq \|g\|_\lambda \leq \|g\|_{L^2(Q)} \quad \text{for all } g \in L^2(Q). \quad (2.12)$$

Assume (P1)–(P3). If $w \in L^2(Q)$, $v \in S_{\Phi, w}^2$, and $u := S_0(v) + w_0$, then for every $\lambda > 0$,

$$\|u\|_\lambda \leq \|w_0\|_\lambda + \frac{\sqrt{2} m_0 \sqrt{|Q|}}{\lambda} + \frac{\sqrt{2} m_0}{\lambda} \|w\|_\lambda.$$

Proof. By (P3), as in the proof of Proposition 2.12,

$$|v(t, x)|^2 \leq 2m_0^2(1 + |w(t, x)|^2) \quad \text{a.e. on } Q.$$

Multiplying by $e^{-2\lambda t} \leq 1$ and integrating over Q gives

$$\|v\|_\lambda \leq \sqrt{2} m_0 (\sqrt{|Q|} + \|w\|_\lambda). \quad (2.13)$$

Let $\tilde{u} := S_0(v)$, so that $\tilde{u}(0) = 0$ and $\partial_t \tilde{u} - \Delta \tilde{u} = v$. Multiplying the energy identity

$$\frac{1}{2} \frac{d}{dt} \|\tilde{u}(t)\|_{L^2(\Omega)}^2 + \|\nabla \tilde{u}(t)\|_{L^2(\Omega)}^2 = \int_{\Omega} v(t, x) \tilde{u}(t, x) dx$$

by $e^{-2\lambda t}$ one has

$$e^{-2\lambda t} \frac{d}{dt} \|\tilde{u}\|_{L^2(\Omega)}^2 = \frac{d}{dt} (e^{-2\lambda t} \|\tilde{u}\|_{L^2(\Omega)}^2) + 2\lambda e^{-2\lambda t} \|\tilde{u}\|_{L^2(\Omega)}^2.$$

Applying Young's inequality,

$$\|v(t)\|_{L^2(\Omega)} \|\tilde{u}(t)\|_{L^2(\Omega)} \leq \frac{\lambda}{2} \|\tilde{u}(t)\|_{L^2(\Omega)}^2 + \frac{1}{2\lambda} \|v(t)\|_{L^2(\Omega)}^2,$$

to the source term, integrating over $(0, T)$, using $\tilde{u}(0) = 0$, and discarding the nonnegative terms $e^{-2\lambda T} \|\tilde{u}(T)\|_{L^2(\Omega)}^2$ and the gradient term, gives $\lambda \|\tilde{u}\|_\lambda^2 \leq \lambda^{-1} \|v\|_\lambda^2$, i.e.

$$\|\tilde{u}\|_\lambda \leq \frac{1}{\lambda} \|v\|_\lambda. \quad (2.14)$$

Combining (2.13)–(2.14) with the triangle inequality $\|u\|_\lambda \leq \|\tilde{u}\|_\lambda + \|w_0\|_\lambda$ yields the claim. $\square$

Fix $\lambda_0 := 2\sqrt{2}m_0 + 1$, so that $\sqrt{2}m_0/\lambda_0 \leq \frac{1}{2}$, and abbreviate $\|\cdot\|_* := \|\cdot\|_{\lambda_0}$. Set

$$R := 2\left(\|w_0\|_* + \frac{\sqrt{2}m_0\sqrt{|Q|}}{\lambda_0}\right), \quad K := \{w \in L^2(Q) : \|w\|_* \leq R\}.$$

The set K is nonempty (it contains 0), closed, bounded, and convex in $L^2(Q)$; by (2.12), every $w \in K$ satisfies $\|w\|_{L^2(Q)} \leq e^{\lambda_0 T} R =: R'$.

Lemma 2.16 (Invariance). $\mathcal{G}(K) \subset K$.

Proof. Let $w \in K$ and $u \in \mathcal{G}(w)$. Then, $\|w\|_* \leq R$, $u = S_0(v) + w_0$ with $v \in S_{\Phi,w}^2$, and due to Lemma 2.15 with $\lambda = \lambda_0$ and the choice of λ_0 ,

$$\|u\|_* \leq \|w_0\|_* + \frac{\sqrt{2}m_0\sqrt{|Q|}}{\lambda_0} + \frac{1}{2}\|w\|_* = \frac{R}{2} + \frac{1}{2}\|w\|_* \leq \frac{R}{2} + \frac{R}{2} = R.$$

Hence $u \in K$. □

Lemma 2.17 (Values of $\mathcal{G}$). *For every $w \in K$, the set $\mathcal{G}(w)$ is nonempty, closed, and convex in $L^2(Q)$.*

Proof. It follows from Proposition 2.12 that $\mathcal{G}(w)$ is nonempty. Indeed, picking any $v \in S_{\Phi,w}^2$ gives $S_0(v) + w_0 \in \mathcal{G}(w)$.

We next turn to the proof of convexity. If $u_i = S_0(v_i) + w_0$ with $v_i \in S_{\Phi,w}^2$, $i = 1, 2$, and $\mu \in [0, 1]$, then, by convexity of $S_{\Phi,w}^2$ (Proposition 2.12) and linearity of S_0 ,

$$\mu u_1 + (1 - \mu)u_2 = S_0(\mu v_1 + (1 - \mu)v_2) + w_0 \in \mathcal{G}(w).$$

Therefore, $\mathcal{G}(w)$ is convex.

Finally, let $\{u_n\}_{n \in \mathbb{N}} \subset \mathcal{G}(w)$ with $u_n \rightarrow u$ strongly in $L^2(Q)$, and write $u_n = S_0(v_n) + w_0$ with $v_n \in S_{\Phi,w}^2$. Since w is fixed, (P3) and (2.10) give a uniform bound $\|v_n\|_{L^2(Q)} \leq C(w)$. By reflexivity of $L^2(Q)$, a subsequence of $\{v_n\}_{n \in \mathbb{N}}$, for which we use the same labeling, satisfies $v_n \rightharpoonup v$ weakly in $L^2(Q)$. By weak continuity of S_0 , $S_0(v_n) \rightharpoonup S_0(v)$ weakly in $L^2(Q)$; since $u_n \rightarrow u$ strongly, uniqueness of weak limits yields $u = S_0(v) + w_0$. Applying Lemma 2.14 with the stationary sequence $w_n \equiv w$ gives $v \in S_{\Phi,w}^2$, whence $u \in \mathcal{G}(w)$. We conclude that $\mathcal{G}(w)$ is closed. □

Lemma 2.18 (Relative compactness of the range). $\overline{\mathcal{G}(K)}$ is a compact subset of $L^2(Q)$.

Proof. Let $u \in \mathcal{G}(w)$ for some $w \in K$. Then $u = S_0(v) + w_0$ with $v \in S_{\Phi,w}^2$ and by (2.10) and estimate $\|w\|_{L^2(Q)} \leq R'$ we have the bound

$$\|v\|_{L^2(Q)}^2 \leq 2m_0^2(|Q| + R'^2) =: M^2,$$

which is uniform over $w \in K$ and over the choice of selection v . Since $S_0 : L^2(Q) \rightarrow \mathcal{W}(0, T)$ is bounded,

$$\|u\|_{\mathcal{W}(0, T)} \leq \|S_0\| M + \|w_0\|_{\mathcal{W}(0, T)},$$

so $\mathcal{G}(K)$ is bounded in $\mathcal{W}(0, T)$ (here $\|S_0\|$ denotes the norm of S_0 in the space of continuous linear operators acting from $L^2(Q)$ to $\mathcal{W}(0, T)$). By the compact embedding $\mathcal{W}(0, T) \hookrightarrow C([0, T]; L^2(\Omega))$ of Lemma 2.10, and hence into $L^2(Q)$, bounded subsets of $\mathcal{W}(0, T)$ are relatively compact in $L^2(Q)$. Thus $\overline{\mathcal{G}(K)}$ is compact in $L^2(Q)$. $\square$

Lemma 2.19 (Sequential closedness of the graph). *If $\{w_n\}_{n \in \mathbb{N}} \subset K$, $u_n \in \mathcal{G}(w_n)$, $w_n \rightarrow w$ and $u_n \rightarrow u$ strongly in $L^2(Q)$, then $u \in \mathcal{G}(w)$.*

Proof. Write $u_n = S_0(v_n) + w_0$ with $v_n \in S_{\Phi, w_n}^2$. By (2.10) and the boundedness of $\{w_n\}_{n \in \mathbb{N}} \subset K$ in $L^2(Q)$, the sequence $\{v_n\}_{n \in \mathbb{N}}$ is bounded in $L^2(Q)$. By reflexivity, a subsequence of $\{v_n\}_{n \in \mathbb{N}}$, for which we use the same labeling, satisfies $v_n \rightharpoonup v$ weakly in $L^2(Q)$. By weak continuity of S_0 , $S_0(v_n) \rightharpoonup S_0(v)$ weakly in $L^2(Q)$; on the other hand $S_0(v_n) = u_n - w_0 \rightarrow u - w_0$ strongly, hence weakly. By uniqueness of weak limits, $u = S_0(v) + w_0$. Since $w_n \rightarrow w$ strongly and $v_n \rightharpoonup v$ weakly, Lemma 2.14 gives $v \in S_{\Phi, w}^2$, hence $u \in \mathcal{G}(w)$. $\square$

Theorem 2.20 (Existence of strong solutions). *Assume (P1)–(P3). Then problem (2.5) admits at least one strong solution, i.e., $\mathcal{S}(u_0) \neq \emptyset$.*

Proof. K is a nonempty closed convex subset of the Banach space $L^2(Q)$. By Lemma 2.17, $\mathcal{G}(w)$ is nonempty, closed, and convex for every $w \in K$, and by Lemma 2.16, $\mathcal{G}(K) \subset K$. By Lemma 2.19, the graph of $\mathcal{G}$ is sequentially closed in $K \times L^2(Q)$, which is hypothesis (a) of Theorem 2.13; by Lemma 2.18, $\mathcal{G}(K)$ is contained in the compact set $\overline{\mathcal{G}(K)} \subset L^2(Q)$, which is hypothesis (b). Theorem 2.13, applied with $X = L^2(Q)$ and $D = K$, yields $w^* \in K$ with $w^* \in \mathcal{G}(w^*)$, i.e. $w^* = S_0(v^*) + w_0$ for some $v^* \in S_{\Phi, w^*}^2$. Setting $u := w^*$, we obtain $u \in \mathcal{W}(0, T)$ with $\partial_t u - \Delta u = v^* \in \Phi(u)$ a.e. on Q and $u(0) = u_0$, so $u \in \mathcal{S}(u_0)$ by Definition 2.11. $\square$

Remark 2.21. Theorem 2.20 asserts existence only. Since Φ is merely upper semicontinuous, $\mathcal{S}(u_0)$ need not be a singleton in general, and the fixed-point argument makes no claim about, and does not need, uniqueness of the selection v^* or of the resulting strong solution. For the relay example of Section 6.3, uniqueness does hold, but for a structural reason unrelated to the fixed-point argument: the equation is a subgradient flow of a convex energy; see Remark 6.1.

Remark 2.22. If Φ is, in addition, *lower* semicontinuous, a Michael-type selection theorem ([11, Lemma 2.1], applied with $D = X = \mathbb{R}$) provides a continuous selection $f(s) \in \Phi(s)$, which by (P3) inherits the linear growth condition $|f(s)| \leq m_0(1 + |s|)$; problem (2.5) then reduces to the semilinear equation $\partial_t u - \Delta u = f(u)$, solvable by a standard Schaefer fixed-point argument. Requiring both upper and lower semicontinuity throughout would, however, make the multifunction continuous and would exclude the relay nonlinearity of Section 6.3, which admits no continuous selection; this is precisely why Theorem 2.20 is formulated under upper semicontinuity alone.

3. A General Distance-Residual Principle

Assume that X and Y are Banach spaces and consider a differential operator $\mathcal{D} : X \rightarrow Y$. Let $\mathcal{F} : X \rightrightarrows Y$ be strict and closed-valued. We consider the abstract set-valued differential problem

$$\mathcal{D}(u) \in \mathcal{F}(u). \quad (3.1)$$

Special cases of (3.1) include differential inclusions $\dot{x}(t) \in F(t, x(t))$ and parabolic partial differential inclusions $\partial_t u - \Delta u \in \Phi(u)$.

Definition 3.1 (Distance residual). Let X and Y be Banach spaces and assume that $\mathcal{D} : X \rightarrow Y$ is a differential operator. Let $\mathcal{F} : X \rightrightarrows Y$ be strict and closed-valued. For any $u \in X$, the *distance residual* of u with respect to (3.1) is defined as

$$R(u) := \text{dist}^2(\mathcal{D}(u), \mathcal{F}(u)) = \inf_{v \in \mathcal{F}(u)} \|\mathcal{D}(u) - v\|_Y^2.$$

In the concrete settings considered below, the elements of X and Y are functions of a variable z ranging over a time interval or a space–time cylinder Z , and the multifunction $\mathcal{F}$ acts through a pointwise set-valued map: there is a multifunction $G : Z \times \mathbb{R}^m \rightrightarrows \mathbb{R}^k$ such that $\mathcal{F}$ is the associated Nemytskii (superposition) operator,

$$\mathcal{F}(u) = \{v \in Y : v(z) \in G(z, u(z)) \text{ for a.e. } z \in Z\},$$

and the inclusion (3.1) is imposed pointwise. In that case we define the *pointwise distance residual*

$$r(u; z) := \text{dist}^2((\mathcal{D}u)(z), G(z, u(z))),$$

and the associated DR-PINN loss is

$$\mathcal{L}(\theta) := \frac{1}{N} \sum_{i=1}^N r(u_\theta; z_i),$$

where u_θ is a neural-network surrogate and $\{z_i\}_{i=1}^N$ are collocation points. Note that $R(u)$ is a single number attached to u , whereas $r(u; \cdot)$ is a function of the collocation variable; the loss $\mathcal{L}$ is an empirical collocation (quadrature) approximation of the continuous residual functional $\int_Z r(u_\theta; z) dz$ (in the experiments below, both deterministic uniform grids and adaptively re-weighted samples are used).

Remark 3.2 (Collocation loss versus continuous functional). The consistency results of Sections 4.3 and 5.2 are formulated for the continuous distance-residual functional $\mathcal{J}$, whereas training minimizes the finite-collocation loss $\mathcal{L}$. An implication passing from $\mathcal{L}(\theta_n) \rightarrow 0$ to $\mathcal{J}(u_{\theta_n}) \rightarrow 0$ requires additional control of the sampling scheme and of the regularity of $r(u_\theta; \cdot)$; it is not part of the consistency theorems. One sufficient bridge is the following. If Z is compact with $|Z| < \infty$, the maps $z \mapsto r(u_{\theta_n}; z)$ share a Lipschitz constant M on Z , the collocation sets $\{z_i^{(n)}\}$ have fill distance $h_n := \sup_{z \in Z} \min_i |z - z_i^{(n)}| \rightarrow 0$, and the *maximum* pointwise residual satisfies $\max_i r(u_{\theta_n}; z_i^{(n)}) \rightarrow 0$, then

$$\sup_{z \in Z} r(u_{\theta_n}; z) \leq \max_i r(u_{\theta_n}; z_i^{(n)}) + M h_n \rightarrow 0,$$

and hence $\int_Z r(u_{\theta_n}; z) dz \rightarrow 0$, as $n \rightarrow \infty$. This condition controls the maximum collocation residual; the mean loss $\mathcal{L}$ alone does not give the same bound. Accordingly, the dense-grid validations in Section 6 report both mean and maximal residuals.

Remark 3.3. Under the hypothesis of the previous definition, the distance residual possesses the following fundamental properties:

- (a) Non-negativity: $R(u) \geq 0$ for all $u \in X$.
- (b) Exactness: if $\mathcal{F}(u)$ is closed, then $R(u) = 0 \iff \mathcal{D}(u) \in \mathcal{F}(u)$.

Moreover, $R(u)^{1/2}$ admits a natural geometric interpretation: it represents the distance from $\mathcal{D}(u)$ to the admissible set $\mathcal{F}(u)$, i.e., the *infimal* correction needed to satisfy the inclusion at u . In a general Banach space the infimum need not be attained; it is attained (and the correction is then a genuine minimum, realized by the metric projection) when $\mathcal{F}(u)$ is closed and convex and Y is a Hilbert space, cf. Section 2.2.

Remark 3.4. Since we are assuming that $\mathcal{F}$ is strict and closed-valued, when $\mathcal{F}(u)$ is convex and Y is a Hilbert space, the projection $P_{\mathcal{F}(u)}(\mathcal{D}(u))$ onto $\mathcal{F}(u)$ is uniquely defined (cf. Section 2.2), and

$$R(u) = \|\mathcal{D}(u) - P_{\mathcal{F}(u)}(\mathcal{D}(u))\|_Y^2.$$

For a *fixed* closed convex set $C \subset Y$, the map $y \mapsto \text{dist}^2(y, C)$ is Fréchet differentiable and

$$\nabla_y \text{dist}^2(y, C) = 2(y - P_C(y)).$$

This projection-gradient identity is used in the numerical implementations. It differentiates with respect to the first argument while holding the set fixed. When $\mathcal{F}(u)$ depends on u , computing $\nabla R(u)$ also requires differentiating that state dependence (or rewriting it explicitly, as in the translated-set examples below).

When $\mathcal{F}(u) = \{f(u)\}$ is a singleton, $R(u) = \|\mathcal{D}(u) - f(u)\|_Y^2$, which is the standard PINN residual.

3.1. A general consistency result

The basic closed-graph argument is the following: sufficiently strong convergence together with a vanishing distance residual forces the limit to satisfy the target inclusion.

Lemma 3.5. *Let X and Y be Banach spaces and let $\mathcal{F} : X \rightrightarrows Y$ be a strict set-valued function with closed graph. Suppose that $x_n \rightarrow x$ in X , $y_n \rightarrow y$ in Y , and*

$$\text{dist}(y_n, \mathcal{F}(x_n)) \rightarrow 0. \tag{3.2}$$

Then $y \in \mathcal{F}(x)$.

Proof. By the definition of the distance to a set, for each $n \in \mathbb{N}$ there exists $a_n \in \mathcal{F}(x_n)$ such that

$$\|y_n - a_n\|_Y \leq \text{dist}(y_n, \mathcal{F}(x_n)) + \frac{1}{n}.$$

From this and (3.2) we deduce that

$$\|y_n - a_n\|_Y \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Consequently, since $y_n \rightarrow y$, we have

$$a_n = y_n - (y_n - a_n) \rightarrow y \quad \text{as } n \rightarrow \infty.$$

Now, since $a_n \in \mathcal{F}(x_n)$, we have $(x_n, a_n) \in \text{gr}(\mathcal{F})$ for all $n \in \mathbb{N}$. Since $\text{gr}(\mathcal{F})$ is closed, $x_n \rightarrow x$, and $a_n \rightarrow y$, it follows that $(x, y) \in \text{gr}(\mathcal{F})$, which gives $y \in \mathcal{F}(x)$. $\square$

Theorem 3.6 (General consistency). *Let X, Y be Banach spaces, assume that $\mathcal{D} : X \rightarrow Y$ is continuous, and let $\mathcal{F} : X \rightrightarrows Y$ be strict and closed (that is, with closed graph). Let $\{u_n\}_{n \in \mathbb{N}} \subset X$ satisfy $u_n \rightarrow u$ in X and*

$$\text{dist}(\mathcal{D}(u_n), \mathcal{F}(u_n)) \rightarrow 0.$$

Then $\mathcal{D}(u) \in \mathcal{F}(u)$.

Proof. Since $\mathcal{D}$ is continuous, $\mathcal{D}(u_n) \rightarrow \mathcal{D}(u)$. The result then follows by applying Lemma 3.5 with $x_n = u_n$ and $y_n = \mathcal{D}(u_n)$. $\square$

Theorem 3.6 requires strong convergence of both u_n and $\mathcal{D}(u_n)$. In the ODE and PDE settings of Sections 4.3 and 5, the differential operators converge only weakly. The corresponding consistency results therefore use compactness of the states together with weak convergence of the operators and lower semicontinuity of the distance-residual functional, in the spirit of the direct method of the calculus of variations.

4. DR-PINNs for Differential Inclusions

4.1. Formulation

Consider the differential inclusion (2.2) with $F : [0, T] \times \mathbb{R}^d \rightrightarrows \mathbb{R}^d$ satisfying the hypotheses of Theorem 2.7. The classical PINN residual

$$\dot{x}(t) - f(t, x(t))$$

has no analogue when F is set-valued and lacks a single-valued selection. The natural replacement is the distance from the computed velocity to the admissible set:

$$R_F(t; \theta) := \text{dist}^2(\dot{x}_\theta(t), F(t, x_\theta(t))), \quad (4.1)$$

where $x_\theta(t)$ is a neural-network surrogate and $\dot{x}_\theta(t)$ is obtained by automatic differentiation, exactly as in classical PINNs.

Remark 4.1. The assumptions that $F(t, x)$ is nonempty and closed ensure that the distance is finite and attained: for every $v \in \mathbb{R}^d$ there exists $\Pi_{F(t, x)}(v) \in F(t, x)$ with

$$\text{dist}(v, F(t, x)) = |v - \Pi_{F(t, x)}(v)|.$$

Closedness gives the following exactness property:

$$\text{dist}(v, F(t, x)) = 0 \iff v \in F(t, x).$$

When $F(t, x)$ is also convex, the projection $\Pi_{F(t, x)}(v)$ is unique (see Section 2.2). The squared distance becomes

$$\text{dist}^2(v, F(t, x)) = |v - \Pi_{F(t, x)}(v)|^2, \quad (4.2)$$

and its gradient with respect to v is

$$\nabla_v \text{dist}^2(v, F(t, x)) = 2(v - \Pi_{F(t, x)}(v)),$$

which is well-defined and Lipschitz. This identity differentiates only with respect to v , with $F(t, x)$ held fixed. If the admissible set depends on the learned state x_θ , the corresponding chain-rule term must be retained in the implementation; exact formulas for the practically relevant cases, and the limits of plain differentiability for general moving sets, are given in Remark 4.2 below.

The DR-PINN loss for the differential inclusion (2.2) is

$$\mathcal{L}_F(\theta) = \frac{1}{N} \sum_{i=1}^N \text{dist}^2(\dot{x}_\theta(t_i), F(t_i, x_\theta(t_i))), \quad (4.3)$$

where $\{t_i\}_{i=1}^N \subset [0, T]$ are collocation points. Boundary and initial conditions are incorporated additively, so that the total loss then takes the form

$$\mathcal{L}(\theta) = \lambda_F \mathcal{L}_F(\theta) + \lambda_0 |x_\theta(0) - x_0|^2,$$

where $\lambda_F, \lambda_0 > 0$ are user-specified weights. Given a neural network $N_\theta(t)$ we set

$$x_\theta(t) = x_0 + t N_\theta(t),$$

which enforces $x_\theta(0) = x_0$, so that only the inclusion residual remains in the loss (4.3), that is, we can take $\mathcal{L}(\theta) = \mathcal{L}_F(\theta)$, which is more convenient for numerical approximations. Alternatively, the initial constraint can be penalized as a distance to a singleton.

4.2. Computing the Projection onto a General Convex Set

To evaluate the loss (4.3) one must compute the projection onto $F(t, x)$. For simple sets (intervals, balls, boxes) this is explicit. For a large and practically relevant class of admissible sets — those that are, or can be approximated by, convex hulls of finitely many vertices — the projection is computable by a small quadratic program, as we now describe.

Suppose that $F(t, x) = \text{conv}(V(t, x))$, where conv denotes the convex hull and

$$V(t, x) = \{v_1(t, x), \dots, v_K(t, x)\} \subset \mathbb{R}^d.$$

This covers any set of the controlled form $F(t, x) = \{f(t, x, u) : u \in U\}$ with U a polytope, *provided* $f(t, x, \cdot)$ is affine in u : the image of a polytope under an affine map is again a polytope, with vertices among the images of the vertices. For a general nonlinear f the image of a polytope need not be a polytope, nor even convex, and the construction below does not apply verbatim. The framework also covers any convex set discretized by a finite point cloud. For every $(t, x) \in [0, T] \times \mathbb{R}^d$, the projection of $v \in \mathbb{R}^d$ onto $F = \text{conv}(V(t, x))$ satisfies

$$\Pi_F(v) = \sum_{k=1}^K \lambda_k^* v_k(t, x), \quad \lambda^* \in \underset{\lambda \in \Delta_K}{\text{argmin}} \left| v - \sum_{k=1}^K \lambda_k v_k(t, x) \right|^2, \quad (4.4)$$

where $\Delta_K = \{\lambda \in \mathbb{R}^K : \lambda_k \geq 0, \sum_k \lambda_k = 1\}$ is the probability simplex. The projected point $\Pi_F(v)$ is unique, since the set is convex, but its barycentric representation λ^* need not be: any minimizer of the quadratic program yields the same projected point. Problem (4.4) is a small convex quadratic program with K variables, one equality constraint, and non-negativity bounds. It can be solved efficiently, even inside the training loop, by standard active-set or Frank–Wolfe-type methods.

This reformulation shifts the difficulty from “compute the projection onto a general convex set” to “produce a vertex set $V(t, x)$ whose convex hull approximates $F(t, x)$ ”. When $V(t, x)$ is redundant, it is advantageous to first reduce it to its extreme points using a convex-hull algorithm such as Quickhull [5]. Quickhull computes the extreme points $\tilde{V}(t, x) \subset V(t, x)$ efficiently in practice (in output-sensitive time in low dimensions [5]) and satisfies $\text{conv}(\tilde{V}(t, x)) = \text{conv}(V(t, x)) = F(t, x)$, while typically having far fewer points, which keeps the quadratic program (4.4) small.

Two regimes are worth distinguishing:

- (a) $F(t, x)$ is constant, or depending only on t . The extreme-point set $\tilde{V}(t)$ can be precomputed *offline* on a grid of times. During training only the small quadratic program (4.4) is solved online.

- (b) $F(t, x)$ depends on the state $x_\theta(t)$. If $V(t, x) = f(t, x) + A(t, x)V_0$ for a fixed reference set V_0 (independent of (t, x)), a convex-hull reduction $\tilde{V}_0$ of V_0 can still be computed once offline. Indeed, for every fixed (t, x) , $A(t, x)$ is simply a linear map, and

$$\begin{aligned}\text{conv}(A(t, x)\tilde{V}_0) &= A(t, x)\text{conv}(\tilde{V}_0) = A(t, x)\text{conv}(V_0) \\ &= \text{conv}(A(t, x)V_0),\end{aligned}$$

an identity valid for *every* linear map. Only the reduced set needs to be transformed — by the current $A(t, x)$ and $f(t, x)$ — at each collocation point during training.

In Section 4.3 below, we impose hypotheses (H1)–(H3) to prove the consistency of the distance residual. What is required so that F satisfies these hypotheses is regularity and growth of $A(\cdot, \cdot)$ and $f(\cdot, \cdot)$. Precisely, $A(\cdot, x)$ and $f(\cdot, x)$ are measurable in t for every x ; $A(t, \cdot)$ and $f(t, \cdot)$ are continuous for a.e. t (which already yields Hausdorff continuity, hence upper semicontinuity, of $x \mapsto F(t, x)$, since $d_H(A_1S, A_2S) \leq \|A_1 - A_2\| \sup_{w \in S} |w|$ for any bounded S); and at most linear growth in x ,

$$|f(t, x)| + \|A(t, x)\| R_0 \leq m(t)(1 + |x|), \quad R_0 := \sup_{w \in V_0} |w|,$$

for some $m \in L^2(0, T)$, so that F inherits the linear-growth bound (H3). Constant $A(t, x)$ (as in Section 6.1, where B plays this role) or $A(t, x)$ affine in x are natural sufficient special cases; a genuinely nonlinear dependence $(t, x) \mapsto A(t, x)$ is equally admissible provided this growth bound holds.

Quickhull is used only as a preprocessing step: it removes redundant points before the quadratic program is solved. Combined with (4.4), this keeps the evaluation of $\text{dist}^2(v, F(t, x))$ practical for convex sets represented by finite point clouds.

Remark 4.2 (Differentiating the residual when the set depends on the state). When the admissible set depends on the learned state, that dependence also enters the derivative. The two cases used in this paper are as follows.

(i) *Translations of a fixed set.* Suppose that

$$F(t, x) = c(t, x) + C(t),$$

where $C(t) \subset \mathbb{R}^d$ is a nonempty closed convex set not depending on x and $c(\cdot, \cdot)$ is C^1 in x . Since the distance is translation invariant,

$$r(t, x, v) := \text{dist}^2(v, c(t, x) + C(t)) = \text{dist}^2(v - c(t, x), C(t)),$$

and the set on the right-hand side no longer moves with x . Writing

$$q := (v - c(t, x)) - \Pi_{C(t)}(v - c(t, x)),$$

the Fréchet differentiability of $y \mapsto \text{dist}^2(y, C(t))$ (whose gradient is $2(y - \Pi_{C(t)}y)$, cf. Section 3) combined with the chain rule gives the *exact* gradients

$$\nabla_v r = 2q, \quad \nabla_x r = -2 D_x c(t, x)^\top q,$$

so r is genuinely C^1 in (x, v) . In an automatic-differentiation implementation, these gradients are obtained by evaluating the projection numerically and holding the *projected point* constant in the backward pass, while differentiating through the argument $v - c(t, x)$. This covers the linear control example of Section 6.1, where $F(t, x) = Ax + BU$, $c(t, x) = Ax$, $D_x c = A$, and the ellipsoidal example of Section 6.2, where $c(t, x) = g(t, x)$.

(ii) *General moving polytopes.* For $F(t, x) = \text{conv}\{v_1(t, x), \dots, v_K(t, x)\}$ with $v_k(t, \cdot) \in C^1$, the residual is the value function of the quadratic program (4.4),

$$r(t, x, v) = \min_{\lambda \in \Delta_K} \left\| v - \sum_{k=1}^K \lambda_k v_k(t, x) \right\|^2.$$

If the minimizer $\lambda^* = \lambda^*(t, x, v)$ is *unique* in a neighborhood of the point of interest, then, by Danskin's theorem for parametric minimization over a compact set, r is differentiable there, with

$$\nabla_v r = 2e, \quad \nabla_x r = -2 \sum_{k=1}^K \lambda_k^* D_x v_k(t, x)^\top e, \quad e := v - \sum_{k=1}^K \lambda_k^* v_k(t, x).$$

If λ^* is *not* unique (the projected point is always unique, but its barycentric representation need not be, cf. the discussion after (4.4)), the value function is in general only directionally differentiable in x , rather than C^1 , so directional derivatives or Clarke subgradients are needed. The plain-gradient statements used here are therefore restricted to fixed sets, sets depending only on t , translations as in (i), and moving polytopes for which the inner quadratic program has a unique, locally stable minimizer.

4.3. Consistency

We next prove the consistency of the distance residual, as stated in Theorem 4.8 below. Since solutions of an inclusion need not be unique, the result is a compactness/lower-semicontinuity statement rather than a Grönwall-type error estimate for a distinguished solution.

We work under the hypotheses of Theorem 2.7, with two mild strengthenings: $m \in L^2(0, T)$, and joint (product) measurability of F in place of measurability in t alone (see Remark 4.3 below):

(H1) F is strict, closed-valued, and convex-valued.

(H2) F is *product measurable* (or jointly $\mathcal{L}([0, T]) \otimes \mathcal{B}(\mathbb{R}^d)$ -measurable), i.e.,

$$\begin{aligned} \{(t, x) \in [0, T] \times \mathbb{R}^d : F(t, x) \cap U \neq \emptyset\} &\in \mathcal{L}([0, T]) \otimes \mathcal{B}(\mathbb{R}^d) \\ &\text{for every open } U \subset \mathbb{R}^d, \end{aligned}$$

where $\mathcal{L}([0, T]) \otimes \mathcal{B}(\mathbb{R}^d)$ is the product σ -algebra of the Lebesgue σ -field $\mathcal{L}([0, T])$ and the Borel σ -field $\mathcal{B}(\mathbb{R}^d)$, and $x \mapsto F(t, x)$ is upper semicontinuous for *a.e.* $t \in [0, T]$.

(H3) There exists $m \in L^2(0, T)$, $m \geq 0$, such that

$$\sup_{v \in F(t, x)} |v| \leq m(t)(1 + |x|) \quad \text{for a.e. } t \in [0, T] \text{ and every } x \in \mathbb{R}^d.$$

Remark 4.3. Fixing $x \in \mathbb{R}^d$ in the product measurability condition in (H2) shows that $t \mapsto F(t, x)$ is Lebesgue measurable for every $x \in \mathbb{R}^d$, hence (H2) implies the measurability hypothesis of Theorem 2.7. Product measurability is a mild requirement: it holds, in particular, whenever F is jointly upper semicontinuous on $[0, T] \times \mathbb{R}^d$ (then $\{F \cap U \neq \emptyset\}$ is an F_σ set for every open U), and whenever F is measurable in t and *continuous* in x (Carathéodory multifunctions); both cases cover all the admissible sets used in Section 6. This strengthening is needed because measurability in t together with upper semicontinuity in x does not, in general, imply superpositional measurability; see [28]. Product measurability in (H2) supplies the measurability used in the proof of Theorem 4.8.

Definition 4.4. Define the admissible space

$$\mathcal{A} := \left\{ x \in W^{1,2}(0, T; \mathbb{R}^d) : x(0) = x_0 \right\},$$

where $W^{1,2}(0, T; \mathbb{R}^d)$ is the Sobolev space of absolutely continuous functions with weak derivative in $L^2(0, T; \mathbb{R}^d)$. For $x \in \mathcal{A}$, define the *distance-residual functional*

$$\mathcal{J}(x) := \int_0^T \text{dist}^2(\dot{x}(t), F(t, x(t))) dt. \quad (4.5)$$

Every trajectory produced by the hard-constraint ansatz $x_\theta(t) = x_0 + tN_\theta(t)$, with N_θ Lipschitz on $[0, T]$ (as for the tanh networks used below), belongs to $\mathcal{A}$. Finite networks represent only a subset of $\mathcal{A}$.

Lemma 4.5. *Let $F : [0, T] \times \mathbb{R}^d \rightrightarrows \mathbb{R}^d$ be a set-valued function satisfying hypothesis (H1) and (H2). Consider the function $g : [0, T] \times \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ defined as*

$$g(t, x, v) := \text{dist}^2(v, F(t, x)).$$

Then, g is a normal convex integrand, that is:

- (a) $g(t, x, \cdot)$ is convex in v for a.e. $t \in [0, T]$ and every $x \in \mathbb{R}^d$.
- (b) $g(t, \cdot, \cdot)$ is jointly lower semicontinuous in (x, v) for a.e. $t \in [0, T]$.
- (c) g is jointly $\mathcal{L}([0, T]) \otimes \mathcal{B}(\mathbb{R}^d \times \mathbb{R}^d)$ -measurable on $[0, T] \times \mathbb{R}^d \times \mathbb{R}^d$.

Proof. For (a), fix $x \in \mathbb{R}^d$. For $v_1, v_2 \in \mathbb{R}^d$ and $\lambda \in [0, 1]$, since $F(t, x)$ is closed and convex, the projections onto $F(t, x)$ exist and are unique:

$$w_1 := \Pi_{F(t, x)}(v_1), \quad w_2 := \Pi_{F(t, x)}(v_2).$$

As $F(t, x)$ is convex, $\bar{w} := \lambda w_1 + (1 - \lambda)w_2 \in F(t, x)$. Consequently,

$$\begin{aligned} \text{dist}^2(\lambda v_1 + (1 - \lambda)v_2, F(t, x)) &\leq |\lambda v_1 + (1 - \lambda)v_2 - \bar{w}|^2 \\ &= |\lambda(v_1 - w_1) + (1 - \lambda)(v_2 - w_2)|^2. \end{aligned} \quad (4.6)$$

On the other hand, since the squared Euclidean norm is convex,

$$|\lambda(v_1 - w_1) + (1 - \lambda)(v_2 - w_2)|^2 \leq \lambda|v_1 - w_1|^2 + (1 - \lambda)|v_2 - w_2|^2. \quad (4.7)$$

Gathering (4.6) and (4.7),

$$\begin{aligned} \text{dist}^2(\lambda v_1 + (1 - \lambda)v_2, F(t, x)) &\leq \lambda|v_1 - w_1|^2 + (1 - \lambda)|v_2 - w_2|^2 \\ &= \lambda \text{dist}^2(v_1, F(t, x)) + (1 - \lambda) \text{dist}^2(v_2, F(t, x)). \end{aligned}$$

We conclude that $g(t, x, \cdot)$ is convex in v .

For (b), we start by proving lower semicontinuity in x , with $v \in \mathbb{R}^d$ fixed. Suppose, for the sake of contradiction, that there is a sequence $\{x_n\}_{n \in \mathbb{N}}$ such that $x_n \rightarrow x \in \mathbb{R}^d$ and

$$\liminf_{n \rightarrow \infty} \text{dist}(v, F(t, x_n)) = L < d := \text{dist}(v, F(t, x)).$$

Choose $d' \in (L, d)$. Since $\text{dist}(v, F(t, x)) = d > d'$, all points in $F(t, x)$ lie outside the closed ball $\overline{B}(v, d')$, that is,

$$F(t, x) \subset U := \mathbb{R}^d \setminus \overline{B}(v, d'),$$

where U is open. By upper semicontinuity of $F(t, \cdot)$ at x (hypothesis (H2)), there exists a neighborhood V of x such that

$$F(t, x') \subset U \quad \text{for all } x' \in V.$$

This means that $|w - v| > d'$ for all $w \in F(t, x')$ and $x' \in V$. Hence,

$$\text{dist}(v, F(t, x')) \geq d' \quad \text{for all } x' \in V. \quad (4.8)$$

Since $x_n \rightarrow x$ and V is a neighborhood of x , there exists $N \geq 1$ such $x_n \in V$ whenever $n > N$. From this and (4.8), we have

$$\text{dist}(v, F(t, x_n)) \geq d' \quad \text{for all } n > N.$$

Hence $\liminf_{n \rightarrow \infty} \text{dist}(v, F(t, x_n)) \geq d' > L$, which contradicts the definition of L and finishes the proof of the lower semicontinuity in x whenever $v \in \mathbb{R}^d$ is fixed.

Next, for a fixed $x \in \mathbb{R}^d$, one can verify that the map $v \mapsto \text{dist}(v, F(t, x))$ is 1-Lipschitz continuous in v , that is,

$$|\text{dist}(v, F(t, x)) - \text{dist}(v', F(t, x))| \leq |v - v'| \quad \text{for each } v, v' \in \mathbb{R}^d. \quad (4.9)$$

Fix $\{x_n\}_{n \in \mathbb{N}} \subset \mathbb{R}^d$ and $\{v_n\}_{n \in \mathbb{N}} \subset \mathbb{R}^d$ such that $(x_n, v_n) \rightarrow (x, v)$ as $n \rightarrow \infty$. From (4.9), we have

$$\text{dist}(v_n, F(t, x_n)) \geq \text{dist}(v, F(t, x_n)) - |v_n - v| \quad \text{for all } n \in \mathbb{N}. \quad (4.10)$$

Taking the $\liminf_{n \rightarrow \infty}$ on both sides, the term $|v_n - v|$ vanishes and the lower semicontinuity in x for the first term on the right-hand side gives

$$\liminf_{n \rightarrow \infty} \text{dist}(v_n, F(t, x_n)) \geq \text{dist}(v, F(t, x)). \quad (4.11)$$

The proof of (b) is completed by squaring both sides.

For (c), fix first $v_0 \in \mathbb{R}^d$ and $c > 0$. Then

$$\begin{aligned} & \{(t, x) : \text{dist}(v_0, F(t, x)) < c\} \\ &= \{(t, x) : F(t, x) \cap B(v_0, c) \neq \emptyset\} \in \mathcal{L}([0, T]) \otimes \mathcal{B}(\mathbb{R}^d) \end{aligned}$$

by the product measurability required in (H2), so $(t, x) \mapsto \text{dist}(v_0, F(t, x))$ is jointly measurable for each fixed v_0 . Combining this with the 1-Lipschitz continuity (4.9) in v and with the exact representation

$$\text{dist}(v, F(t, x)) = \inf_{q \in \mathbb{Q}^d} [\text{dist}(q, F(t, x)) + |v - q|]$$

(the inequality “ $\leq$ ” follows from the triangle inequality, and “ $\geq$ ” follows by letting $q \rightarrow v$), we express $(t, x, v) \mapsto \text{dist}(v, F(t, x))$ as a countable infimum of jointly measurable functions. Hence $g(t, x, v) = \text{dist}^2(v, F(t, x))$ is jointly measurable, which proves (c). $\square$

The next lemma settles the measurability of the superposed multifunction and of the associated metric projection appearing in the proof of Theorem 4.8.

Lemma 4.6 (Measurability of superpositions and projections). *Let $F : [0, T] \times \mathbb{R}^d \rightrightarrows \mathbb{R}^d$ satisfy (H1)–(H2). Then:*

- (a) *For every Lebesgue measurable $y : [0, T] \rightarrow \mathbb{R}^d$, the multifunction $t \mapsto F(t, y(t))$ is Lebesgue measurable, strict, closed-valued, and convex-valued.*
- (b) *If, in addition, $v : [0, T] \rightarrow \mathbb{R}^d$ is measurable, then the metric projection $t \mapsto \Pi_{F(t, y(t))}(v(t))$ is measurable.*

Proof. (a) The map $t \mapsto (t, y(t))$ is measurable from $([0, T], \mathcal{L}([0, T]))$ into $([0, T] \times \mathbb{R}^d, \mathcal{L}([0, T]) \otimes \mathcal{B}(\mathbb{R}^d))$, since the preimage of every product set $A \times B$ is $A \cap y^{-1}(B) \in \mathcal{L}([0, T])$ and such products generate the product σ -field. Hence, for every open $U \subset \mathbb{R}^d$,

$$\{t \in [0, T] : F(t, y(t)) \cap U \neq \emptyset\} = \{t : (t, y(t)) \in \{F \cap U \neq \emptyset\}\} \in \mathcal{L}([0, T])$$

by the product measurability in (H2). The values are nonempty, closed, and convex by (H1).

(b) Since $t \mapsto F(t, y(t))$ is measurable with closed convex values and $v(\cdot)$ is measurable, the metric projection is measurable by the measurable-projection theorem [3, Theorem 8.2.11]. $\square$

Both consistency proofs use the same lower-semicontinuity principle: strong convergence in the state variable together with weak convergence in the variable in which the integrand is convex.

Lemma 4.7 (Strong–weak lower semicontinuity). *Let $Z \subset \mathbb{R}^N$ be bounded and measurable, and let $g : Z \times \mathbb{R}^m \times \mathbb{R}^k \rightarrow [0, \infty]$ be a normal integrand, i.e., jointly $\mathcal{L}(Z) \otimes \mathcal{B}(\mathbb{R}^m \times \mathbb{R}^k)$ -measurable and such that $g(z, \cdot, \cdot)$ is lower semicontinuous on $\mathbb{R}^m \times \mathbb{R}^k$ for a.e. $z \in Z$. Assume in addition that $g(z, x, \cdot)$ is convex for a.e. $z \in Z$ and every $x \in \mathbb{R}^m$ and that*

$$\int_Z g(z, x_0(z), v_0(z)) \, dz < \infty$$

for some $x_0 \in L^2(Z; \mathbb{R}^m)$, $v_0 \in L^2(Z; \mathbb{R}^k)$. If

$$x_n \rightarrow x \text{ strongly in } L^2(Z; \mathbb{R}^m), \quad v_n \rightharpoonup v \text{ weakly in } L^2(Z; \mathbb{R}^k),$$

then

$$\int_Z g(z, x(z), v(z)) \, dz \leq \liminf_{n \rightarrow \infty} \int_Z g(z, x_n(z), v_n(z)) \, dz.$$

Proof. The result follows directly from [16, Theorem 3] since g is nonnegative. $\square$

Theorem 4.8 (Consistency of the distance residual). *Let $F : [0, T] \times \mathbb{R}^d \rightrightarrows \mathbb{R}^d$ be a set-valued function satisfying hypotheses (H1)–(H3). Consider the differential inclusion (2.2), along with its solution set $\mathcal{S}(x_0)$ defined as in (2.4). Then:*

- (a) (Exactness) *For every $x \in \mathcal{A}$, $\mathcal{J}(x) = 0$ if and only if $x \in \mathcal{S}(x_0)$.*
- (b) (Asymptotic consistency) *If $\{x_n\}_{n \in \mathbb{N}} \subset \mathcal{A}$ satisfies $\mathcal{J}(x_n) \rightarrow 0$, then there exists a subsequence of $\{x_n\}_{n \in \mathbb{N}}$, for which we use the same labeling, and $x^* \in \mathcal{A}$ such that*

$$x_n \rightarrow x^* \text{ uniformly on } [0, T], \quad \dot{x}_n \rightharpoonup \dot{x}^* \text{ weakly in } L^2(0, T; \mathbb{R}^d),$$

and

$$x^* \in \mathcal{S}(x_0).$$

Proof. Part (a) follows immediately from the exactness property of the distance. To prove (b) we start by setting $r_n(t) := \text{dist}(\dot{x}_n(t), F(t, x_n(t)))$ for every $n \in \mathbb{N}$ so that

$$\int_0^T r_n^2(t) \, dt = \mathcal{J}(x_n) =: \varepsilon_n \rightarrow 0.$$

Let $\varepsilon_0 := \sup_n \varepsilon_n < \infty$. Since $F(t, x_n(t))$ is nonempty, closed, and convex, the projection

$$w_n(t) = \Pi_{F(t, x_n(t))}(\dot{x}_n(t)) \in F(t, x_n(t))$$

is uniquely defined and the minimum is attained, that is,

$$|\dot{x}_n(t) - w_n(t)| = \min_{w \in F(t, x_n(t))} |\dot{x}_n(t) - w| = r_n(t).$$

Moreover, $t \mapsto w_n(t)$ is (Lebesgue) measurable by Lemma 4.6, applied with $y = x_n$ and $v = \dot{x}_n$. By the triangle inequality and (H3), for a.e. $t \in [0, T]$,

$$|\dot{x}_n(t)| \leq |\dot{x}_n(t) - w_n(t)| + |w_n(t)| = r_n(t) + m(t)(1 + |x_n(t)|). \quad (4.12)$$

On the other hand, applying Cauchy–Schwarz inequality to $r_n(t)$, we obtain

$$\int_0^t r_n(s) ds \leq \left(\int_0^T ds \right)^{1/2} \left(\int_0^T r_n^2(s) ds \right)^{1/2} \leq \sqrt{T\varepsilon_n} \leq \sqrt{T\varepsilon_0} =: R_0.$$

Then, integrating (4.12) we deduce that

$$\begin{aligned} |x_n(t)| &\leq |x_0| + |x_n(t) - x_0| \leq |x_0| + \int_0^t |\dot{x}_n(s)| ds \\ &\leq |x_0| + R_0 + \int_0^t m(s)(1 + |x_n(s)|) ds. \end{aligned}$$

From this and Grönwall’s inequality (see, e.g., [? , pp. 12–14]) we obtain

$$1 + |x_n(t)| \leq (1 + |x_0| + R_0)e^{\|m\|_{L^1(0,T)}} := C_0 \quad (4.13)$$

for every $t \in [0, T]$ and $n \in \mathbb{N}$. Substituting (4.13) back into (4.12), squaring, and integrating,

$$\int_0^T |\dot{x}_n(t)|^2 dt \leq \int_0^T r_n^2(t) dt + 2C_0 \int_0^T r_n(t)m(t) dt + C_0^2 \int_0^T m^2(t) dt.$$

The second term in the right-hand side can be handled by applying Cauchy–Schwarz to obtain

$$\int_0^T r_n(t)m(t) dt \leq \left(\int_0^T r_n^2(t) dt \right)^{1/2} \cdot \left(\int_0^T m^2(t) dt \right)^{1/2} = \sqrt{\varepsilon_n} \|m\|_{L^2(0,T)}.$$

Therefore,

$$\int_0^T |\dot{x}_n(t)|^2 dt \leq \varepsilon_n + 2C_0\sqrt{\varepsilon_n} \|m\|_{L^2(0,T)} + C_0^2 \|m\|_{L^2(0,T)}^2.$$

Since $\varepsilon_0 := \sup_n \varepsilon_n < \infty$, for every $n \in \mathbb{N}$,

$$\|\dot{x}_n\|_{L^2(0,T)}^2 \leq \varepsilon_0 + 2C_0\sqrt{\varepsilon_0}\|m\|_{L^2(0,T)} + C_0^2\|m\|_{L^2(0,T)}^2 =: C_1. \quad (4.14)$$

This shows that the sequence $\{\dot{x}_n\}_{n \in \mathbb{N}}$ is bounded in $L^2(0, T; \mathbb{R}^d)$. In particular, from (4.13) and (4.14) we deduce that the sequence $\{x_n\}_{n \in \mathbb{N}}$ is bounded in the Sobolev space $W^{1,2}(0, T; \mathbb{R}^d)$.

Since, by (4.14), the sequence $\{\dot{x}_n\}_{n \in \mathbb{N}}$ is bounded in the Hilbert space $L^2(0, T; \mathbb{R}^d)$, which is reflexive, [9, Theorem 3.18] provides a subsequence of $\{\dot{x}_n\}_{n \in \mathbb{N}}$, for which we use the same labeling, and $v^* \in L^2(0, T; \mathbb{R}^d)$ such that

$$\dot{x}_n \rightharpoonup v^* \quad \text{weakly in } L^2(0, T; \mathbb{R}^d).$$

By Cauchy–Schwarz inequality,

$$|x_n(t) - x_n(s)| = \left| \int_s^t \dot{x}_n(\tau) d\tau \right| \leq \|\dot{x}_n\|_{L^2(s,t)} \sqrt{|t-s|} \leq \|\dot{x}_n\|_{L^2(0,T)} \sqrt{|t-s|}.$$

From this and (4.14), we have

$$|x_n(t) - x_n(s)| \leq \sqrt{C_1|t-s|},$$

hence x_n is uniformly equicontinuous. Since x_n is also uniformly bounded as seen from (4.13), we may therefore apply the Arzelà–Ascoli theorem, which guarantees the existence of a subsequence, still denoted by $\{x_n\}_{n \in \mathbb{N}}$, such that

$$x_n \rightarrow x^* \quad \text{uniformly on } [0, T].$$

Then, passing to the limit in $x_n(t) = x_0 + \int_0^t \dot{x}_n(s) ds$ (the integral converges by the weak convergence $\dot{x}_n \rightharpoonup v^*$ in L^2 , tested against $\mathbf{1}_{[0,t]}$) gives

$$x^*(t) = x_0 + \int_0^t v^*(s) ds$$

hence $x^* \in \mathcal{A}$ and $\dot{x}^* = v^*$ a.e.

Finally, set $g(t, x, v) := \text{dist}^2(v, F(t, x))$. By Lemma 4.5, g is a normal convex integrand. The finiteness hypothesis of Lemma 4.7 is satisfied as well: by (H1) and (H3), picking any element of the nonempty set $F(t, 0)$ gives $\text{dist}(0, F(t, 0)) \leq m(t)$ for a.e. t , so the constant functions $x_0 \equiv 0$, $v_0 \equiv 0$ yield $\int_0^T \text{dist}^2(0, F(t, 0)) dt \leq \|m\|_{L^2(0,T)}^2 < \infty$. Since $x_n \rightarrow x^*$ uniformly on $[0, T]$, in particular $x_n \rightarrow x^*$ strongly in $L^2(0, T; \mathbb{R}^d)$, while $\dot{x}_n \rightharpoonup \dot{x}^*$ weakly in $L^2(0, T; \mathbb{R}^d)$. Lemma 4.7, applied with $Z = (0, T)$ and $m = k = d$, yields

$$\mathcal{J}(x^*) \leq \liminf_{n \rightarrow \infty} \mathcal{J}(x_n) = 0.$$

As $\mathcal{J}$ is non-negative, we conclude that $\mathcal{J}(x^*) = 0$. By part (a), this means that $x^* \in \mathcal{S}(x_0)$. $\square$

Remark 4.9. Because solutions need not be unique, Theorem 4.8 identifies a subsequential limit in $\mathcal{S}(x_0)$ but does not select a prescribed element of that set. Additional structure, such as monotonicity, a one-sided Lipschitz condition, or uniqueness, may yield full-sequence convergence, stronger convergence, or quantitative error estimates.

5. DR-PINNs for Partial Differential Inclusions

Consider now the parabolic partial differential inclusion (2.5).

5.1. The Distance-Based PDE Loss

The classical PDE residual $\partial_t u_\theta - \Delta u_\theta - f(u_\theta)$ has no analogue when Φ is set-valued. The natural replacement is

$$\mathcal{L}_{\text{incl}}(\theta) = \frac{1}{N_r} \sum_{i=1}^{N_r} \text{dist}^2\left(\partial_t u_\theta(t_i, x_i) - \Delta u_\theta(t_i, x_i), \Phi(u_\theta(t_i, x_i))\right), \quad (5.1)$$

where $\{(t_i, x_i)\}_{i=1}^{N_r}$ are collocation points in Q . As in the case of differential inclusions, $\mathcal{L}_{\text{incl}}(\theta) = 0$ if and only if the inclusion is satisfied at every collocation point. The full DR-PINN loss incorporates initial and boundary data:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{incl}}(\theta) + \lambda_0 \mathcal{L}_0(\theta) + \lambda_b \mathcal{L}_b(\theta),$$

where

$$\mathcal{L}_0(\theta) = \frac{1}{N_0} \sum_{j=1}^{N_0} |u_\theta(0, x_j^0) - u_0(x_j^0)|^2, \quad \mathcal{L}_b(\theta) = \frac{1}{N_b} \sum_{k=1}^{N_b} |u_\theta(t_k^b, x_k^b)|^2.$$

5.2. Consistency

For the consistency analysis, we assume that the set-valued reaction term $\Phi : \mathbb{R} \rightrightarrows \mathbb{R}$ satisfies the standing hypotheses (P1)–(P3) of Section 2.4; these are exactly the assumptions under which Theorem 2.20 guarantees $\mathcal{S}(u_0) \neq \emptyset$, so that existence and consistency rely on the same set of hypotheses.

Definition 5.1. Let

$$\mathcal{A} := \{u \in \mathcal{W}(0, T) : u(0, \cdot) = u_0 \text{ in } \Omega, u(t, \cdot)|_{\partial\Omega} = 0 \ \forall t \in [0, T]\}.$$

For $u \in \mathcal{A}$, define the *continuous distance-residual functional*

$$\mathcal{J}(u) := \iint_Q \text{dist}^2(\partial_t u - \Delta u, \Phi(u)) \, dx \, dt.$$

The pointwise-in-time boundary condition in the definition of $\mathcal{A}$ is in fact automatically satisfied: by the continuous embedding (2.6), every $u \in \mathcal{W}(0, T)$ admits a representative in $C([0, T]; H_0^1(\Omega))$, so $u(t, \cdot) \in H_0^1(\Omega)$ — and hence $u(t, \cdot)|_{\partial\Omega} = 0$ in the trace sense — for every $t \in [0, T]$, even though $u(t, \cdot) \in H^2(\Omega)$ only for a.e. t . Consequently,

$$\mathcal{A} = \{u \in \mathcal{W}(0, T) : u(0, \cdot) = u_0\}.$$

The boundary condition is retained in the definition only for emphasis.

We also note that $\mathcal{J}(u) < \infty$ for every $u \in \mathcal{A}$: by (P3), choosing for a.e. (t, x) any $r_0(t, x) \in \Phi(u(t, x))$ gives

$$\text{dist}(\partial_t u - \Delta u, \Phi(u)) \leq |\partial_t u - \Delta u - r_0| \leq |\partial_t u - \Delta u| + m_0(1 + |u|) \quad \text{a.e. on } Q;$$

squaring and integrating, and using $\partial_t u - \Delta u \in L^2(Q)$ and $u \in L^2(Q)$ (both immediate from $u \in \mathcal{W}(0, T)$), yields $\mathcal{J}(u) < \infty$.

Lemma 5.2 (Normal convex integrand). *Let $\Phi : \mathbb{R} \rightrightarrows \mathbb{R}$ satisfy hypotheses (P1) and (P2). Define $g : \mathbb{R} \times \mathbb{R} \rightarrow [0, \infty)$ by*

$$g(s, z) := \text{dist}^2(z, \Phi(s)).$$

Then g is a normal convex integrand, that is:

- (a) $g(s, \cdot)$ is convex for every $s \in \mathbb{R}$;
- (b) g is jointly lower semicontinuous on $\mathbb{R} \times \mathbb{R}$;
- (c) g is Borel measurable; consequently, for every pair of measurable functions $u, v : Q \rightarrow \mathbb{R}$, the composition $(t, x) \mapsto g(u(t, x), v(t, x))$ is Lebesgue measurable on Q .

Proof. We treat the three assertions in order.

Part (a). Fix $s \in \mathbb{R}$. Since $\Phi(s)$ is nonempty, closed, and convex by (P1), the projection $\Pi_{\Phi(s)}$ is well defined and single-valued (see Section 2.2). For

$z_1, z_2 \in \mathbb{R}$ and $\lambda \in [0, 1]$, set $w_1 := \Pi_{\Phi(s)}(z_1)$, $w_2 := \Pi_{\Phi(s)}(z_2)$. By convexity of $\Phi(s)$, $w := \lambda w_1 + (1 - \lambda)w_2 \in \Phi(s)$, hence

$$\begin{aligned} \text{dist}^2(\lambda z_1 + (1 - \lambda)z_2, \Phi(s)) &\leq |\lambda z_1 + (1 - \lambda)z_2 - w|^2 \\ &= |\lambda(z_1 - w_1) + (1 - \lambda)(z_2 - w_2)|^2. \end{aligned}$$

By convexity of $t \mapsto t^2$,

$$|\lambda(z_1 - w_1) + (1 - \lambda)(z_2 - w_2)|^2 \leq \lambda(z_1 - w_1)^2 + (1 - \lambda)(z_2 - w_2)^2,$$

and combining the two inequalities gives

$$g(s, \lambda z_1 + (1 - \lambda)z_2) \leq \lambda g(s, z_1) + (1 - \lambda)g(s, z_2).$$

Part (b). We first show that $s \mapsto \text{dist}(z, \Phi(s))$ is lower semicontinuous for fixed $z \in \mathbb{R}$. Suppose, for contradiction, that $s_n \rightarrow s$ and

$$\liminf_{n \rightarrow \infty} \text{dist}(z, \Phi(s_n)) = L < d := \text{dist}(z, \Phi(s)).$$

Choose $d' \in (L, d)$. Since $d > d'$, we have $\Phi(s) \subset U := \mathbb{R} \setminus [z - d', z + d']$, an open set. By the upper semicontinuity of Φ at s (hypothesis (P2)), there exists a neighborhood V of s such that $\Phi(s') \subset U$ for all $s' \in V$. Hence $|w - z| > d'$ for every $w \in \Phi(s')$, $s' \in V$, so that

$$\text{dist}(z, \Phi(s')) \geq d' \quad \text{for all } s' \in V.$$

Since $s_n \rightarrow s$, there exists $N \in \mathbb{N}$ such that $s_n \in V$ for all $n > N$, whence $\text{dist}(z, \Phi(s_n)) \geq d'$ for $n > N$. This contradicts $\liminf_n \text{dist}(z, \Phi(s_n)) = L < d'$, proving lower semicontinuity in s .

On the other hand, for fixed $s \in \mathbb{R}$, the map $z \mapsto \text{dist}(z, \Phi(s))$ is 1-Lipschitz:

$$|\text{dist}(z, \Phi(s)) - \text{dist}(z', \Phi(s))| \leq |z - z'| \quad \text{for all } z, z' \in \mathbb{R}.$$

Let $(s_n, z_n) \rightarrow (s, z)$. Then

$$\text{dist}(z_n, \Phi(s_n)) \geq \text{dist}(z, \Phi(s_n)) - |z_n - z|.$$

Taking $\liminf_{n \rightarrow \infty}$ on both sides, the term $|z_n - z|$ vanishes and the lower semicontinuity in s established above gives

$$\liminf_{n \rightarrow \infty} \text{dist}(z_n, \Phi(s_n)) \geq \text{dist}(z, \Phi(s)).$$

Squaring both (nonnegative) sides yields $\liminf_n g(s_n, z_n) \geq g(s, z)$, which proves (b).

Part (c). A jointly lower semicontinuous function on a metric space has closed sublevel sets $\{(s, z) : g(s, z) \leq c\}$ for every $c \in \mathbb{R}$, hence g is Borel measurable on $\mathbb{R} \times \mathbb{R}$. If $u, v : Q \rightarrow \mathbb{R}$ are Lebesgue measurable, then $(t, x) \mapsto (u(t, x), v(t, x))$ is measurable from Q into $\mathbb{R}^2$, and the composition of a Borel measurable function with a Lebesgue measurable function is Lebesgue measurable. Therefore $(t, x) \mapsto g(u(t, x), v(t, x))$ is measurable on Q . $\square$

Theorem 5.3 (Consistency: parabolic case). *Assume that $\Phi : \mathbb{R} \rightrightarrows \mathbb{R}$ satisfies (P1)–(P3). Then:*

(a) Exactness. *For every $u \in \mathcal{A}$,*

$$\mathcal{J}(u) = 0 \iff \partial_t u(t, x) - \Delta u(t, x) \in \Phi(u(t, x)) \quad \text{for a.e. } (t, x) \in Q.$$

(b) Asymptotic consistency. *Let $\{u_n\}_{n \in \mathbb{N}} \subset \mathcal{A}$ satisfy*

$$\mathcal{J}(u_n) \longrightarrow 0.$$

Then there exists a subsequence, for which we use the same labeling, and $u^ \in \mathcal{A}$ such that*

$$u_n \rightharpoonup u^* \text{ weakly in } \mathcal{W}(0, T), \quad u_n \longrightarrow u^* \text{ strongly in } C([0, T]; L^2(\Omega)),$$

and

$$u^* \in \mathcal{S}(u_0).$$

Proof. Part (a) follows directly from the fact that $\Phi(u(t, x))$ is closed for almost every $(t, x) \in Q$.

We now prove part (b). Set

$$z_n := \partial_t u_n - \Delta u_n$$

and

$$\rho_n(t, x) := \text{dist}(z_n(t, x), \Phi(u_n(t, x))).$$

Then

$$\|\rho_n\|_{L^2(Q)}^2 = \mathcal{J}(u_n) \longrightarrow 0. \tag{5.2}$$

By (P1), the set $\Phi(s)$ is nonempty, closed, and convex for every $s \in \mathbb{R}$. Hence the metric projection onto $\Phi(s)$ is well defined and unique. Define

$$w_n(t, x) := \Pi_{\Phi(u_n(t, x))}(z_n(t, x)).$$

The map w_n is (Lebesgue) measurable on Q . Indeed, by (P2), for every closed $C \subset \mathbb{R}$ the set $\{s \in \mathbb{R} : \Phi(s) \subset \mathbb{R} \setminus C\}$ is open, so $\{s : \Phi(s) \cap C \neq \emptyset\}$ is closed; writing an open set $U \subset \mathbb{R}$ as a countable union of closed sets shows that $\{s : \Phi(s) \cap U \neq \emptyset\}$ is Borel for every open U , i.e., Φ is a Borel measurable multifunction with closed convex values. Consequently, $(t, x) \mapsto \Phi(u_n(t, x))$ is Lebesgue measurable on Q , as the composition of a Borel measurable multifunction with the measurable function u_n , and the metric projection of the measurable function z_n onto it is measurable by the measurable-projection theorem for measurable, closed- and convex-valued multifunctions; see [3, Theorem 8.2.11]. It satisfies

$$w_n(t, x) \in \Phi(u_n(t, x)) \quad \text{for a.e. } (t, x) \in Q,$$

together with

$$|z_n(t, x) - w_n(t, x)| = \rho_n(t, x).$$

Setting

$$r_n := z_n - w_n,$$

we obtain

$$\partial_t u_n - \Delta u_n = w_n + r_n \quad \text{on } Q, \quad (5.3)$$

and, by (5.2),

$$\|r_n\|_{L^2(Q)}^2 = \mathcal{J}(u_n) \longrightarrow 0. \quad (5.4)$$

We first derive an L^2 energy estimate. Testing (5.3) with $u_n(t)$ and using the homogeneous Dirichlet boundary condition, we obtain, for almost every $t \in (0, T)$,

$$\frac{1}{2} \frac{d}{dt} \|u_n(t)\|_{L^2(\Omega)}^2 + \|\nabla u_n(t)\|_{L^2(\Omega)}^2 = \int_{\Omega} (w_n(t, x) + r_n(t, x)) u_n(t, x) dx. \quad (5.5)$$

By (P3),

$$|w_n(t, x)| \leq m_0(1 + |u_n(t, x)|) \quad \text{for a.e. } (t, x) \in Q. \quad (5.6)$$

Consequently,

$$\|w_n(t)\|_{L^2(\Omega)} \leq m_0 \left(|\Omega|^{1/2} + \|u_n(t)\|_{L^2(\Omega)} \right). \quad (5.7)$$

Using the Cauchy–Schwarz and Young inequalities in (5.5), together with (5.7), we find a constant $C > 0$, independent of n , such that

$$\frac{d}{dt} \|u_n(t)\|_{L^2(\Omega)}^2 + 2\|\nabla u_n(t)\|_{L^2(\Omega)}^2 \leq C \left(1 + \|u_n(t)\|_{L^2(\Omega)}^2 \right) + \|r_n(t)\|_{L^2(\Omega)}^2. \quad (5.8)$$

Since $u_n(0) = u_0$ for every n , Gronwall's inequality and (5.4) yield

$$\sup_{n \in \mathbb{N}} \|u_n\|_{L^\infty(0,T;L^2(\Omega))} < \infty. \quad (5.9)$$

Integrating (5.8) over $(0, T)$ also gives

$$\sup_{n \in \mathbb{N}} \|u_n\|_{L^2(0,T;H_0^1(\Omega))} < \infty.$$

From (5.6) and (5.9), it follows that

$$\sup_{n \in \mathbb{N}} \|w_n\|_{L^2(Q)} < \infty.$$

Together with (5.4), this implies

$$\sup_{n \in \mathbb{N}} \|w_n + r_n\|_{L^2(Q)} < \infty. \quad (5.10)$$

We now apply the standard L^2 maximal-regularity estimate for the Dirichlet heat equation to (5.3); see [12, Section 7.1.3, Theorem 5] for smooth domains, with the underlying H^2 elliptic estimate supplied by [14, Theorem 3.2.1.2] on convex domains and by [14, Theorem 2.4.2.5] on $C^{1,1}$ domains. Since $u_0 \in H_0^1(\Omega)$ and $\partial\Omega$ is $C^{1,1}$ or Ω is convex (which covers the square domain used in Section 6.3), there exists a constant $C_{\text{par}} > 0$, independent of n , such that

$$\|u_n\|_{\mathcal{W}(0,T)} \leq C_{\text{par}} \left(\|w_n + r_n\|_{L^2(Q)} + \|u_0\|_{H_0^1(\Omega)} \right).$$

By (5.10),

$$\sup_{n \in \mathbb{N}} \|u_n\|_{\mathcal{W}(0,T)} < \infty.$$

Since $\mathcal{W}(0, T)$ is reflexive, there exist a subsequence, not relabeled, and $u^* \in \mathcal{W}(0, T)$ such that

$$u_n \rightharpoonup u^* \quad \text{weakly in } \mathcal{W}(0, T). \quad (5.11)$$

By the compact parabolic embedding $\mathcal{W}(0, T) \hookrightarrow C([0, T]; L^2(\Omega))$ of Lemma 2.10,

$$u_n \longrightarrow u^* \quad \text{strongly in } C([0, T]; L^2(\Omega)). \quad (5.12)$$

In particular,

$$u_n \longrightarrow u^* \quad \text{strongly in } L^2(Q). \quad (5.13)$$

The convergence in (5.12) and the identity $u_n(0) = u_0$ imply

$$u^*(0) = u_0.$$

Moreover, since

$$u_n \in L^2(0, T; H^2(\Omega) \cap H_0^1(\Omega))$$

and the latter is a closed linear subspace, the weak limit satisfies the homogeneous Dirichlet boundary condition. Hence $u^* \in \mathcal{A}$.

Define

$$g : \mathbb{R} \times \mathbb{R} \longrightarrow [0, \infty), \quad g(s, z) := \text{dist}^2(z, \Phi(s)).$$

The linear operator

$$\mathcal{D} : \mathcal{W}(0, T) \longrightarrow L^2(Q), \quad \mathcal{D}u := \partial_t u - \Delta u,$$

is continuous. Therefore, (5.11) implies

$$\partial_t u_n - \Delta u_n \rightharpoonup \partial_t u^* - \Delta u^* \quad \text{weakly in } L^2(Q). \quad (5.14)$$

By Lemma 5.2, g is Borel measurable, jointly lower semicontinuous, and convex in its second argument. Hence $\tilde{g}(z, s, \zeta) := g(s, \zeta)$, regarded as an integrand on $Q \times \mathbb{R} \times \mathbb{R}$ independent of $z = (t, x)$, satisfies the hypotheses of Lemma 4.7 with $Z = Q$ and $m = k = 1$; in particular, its finiteness hypothesis holds because $\mathcal{J}(u) < \infty$ for every $u \in \mathcal{A}$, as verified after Definition 5.1. Applying that lemma with (5.13) ($u_n \rightarrow u^*$ strongly in $L^2(Q)$) and (5.14) ($\partial_t u_n - \Delta u_n \rightharpoonup \partial_t u^* - \Delta u^*$ weakly in $L^2(Q)$), we obtain

$$\begin{aligned} \mathcal{J}(u^*) &= \int_Q g(u^*, \partial_t u^* - \Delta u^*) \, dx \, dt \\ &\leq \liminf_{n \rightarrow \infty} \int_Q g(u_n, \partial_t u_n - \Delta u_n) \, dx \, dt \\ &= \liminf_{n \rightarrow \infty} \mathcal{J}(u_n) = 0. \end{aligned}$$

Since $\mathcal{J}$ is nonnegative, $\mathcal{J}(u^*) = 0$.

Part (a) then gives

$$\partial_t u^* - \Delta u^* \in \Phi(u^*) \quad \text{for a.e. } (t, x) \in Q.$$

Thus u^* is a strong solution of (2.5), and consequently $u^* \in \mathcal{S}(u_0)$. $\square$

6. Numerical Simulations

We consider three numerical examples: two differential inclusions and one parabolic partial differential inclusion.

6.1. Linear control system with polytopic input set

Problem formulation. We consider the two-dimensional linear differential inclusion

$$\dot{x}(t) \in Ax(t) + BU, \quad x(0) = x_0, \quad t \in [0, T], \quad (6.1)$$

where the admissible-velocity set is

$$F(t, x) = \text{conv}(V(x)), \quad V(x) = \{Ax + Bu_k\}_{k=1}^K. \quad (6.2)$$

This is an instance of regime (b) in Section 4.2: the vertex set $V(x)$ depends on the current state through an affine image of the fixed polytope $U = \text{conv}\{u_1, \dots, u_K\}$.

System data. The state dimension is $d = 2$ and the control dimension is $m = 2$. The drift matrix and the control-influence matrix are

$$A = \begin{pmatrix} -0.5 & 1.0 \\ -1.0 & -0.5 \end{pmatrix}, \quad B = I_2, \quad (6.3)$$

where I_2 denotes the 2×2 identity matrix. The eigenvalues of A are $-0.5 \pm i$, so the uncontrolled system is a stable spiral. The initial condition is $x_0 = (1, 0)^\top$ and the time horizon is $T = 1.5$.

Control set and Quickhull preprocessing. The control set is the ℓ^1 diamond in $\mathbb{R}^2$,

$$U = \text{conv}\{e_1, -e_1, e_2, -e_2\}, \quad (6.4)$$

where e_1, e_2 are the standard basis vectors. To illustrate the computational benefit of the Quickhull reduction, we augment the four true extreme vertices with eight additional points drawn uniformly from $[-0.35, 0.35]^2$, which lie in the interior of $\text{conv}(U)$. The resulting raw vertex set has $K_{\text{raw}} = 12$ points. Applying the Quickhull algorithm (Section 4.2, regime (b)) to BU_{raw} recovers the four true extreme points, reducing the QP (4.4) from $K_{\text{raw}} = 12$ to $\tilde{K} = 4$ variables (see Figure 1).

Since $B = I_2$, the offline-reduced set is $\tilde{U} = \{e_1, -e_1, e_2, -e_2\}$, and the vertex set at each collocation point is built online as

$$\tilde{V}(x_\theta(t_i)) = \{Ax_\theta(t_i) + \tilde{u}_k\}_{k=1}^{\tilde{K}}. \quad (6.5)$$

The projection $\Pi_{F_i}(\dot{x}_\theta(t_i))$ is then computed by solving the QP (4.4) with $\tilde{K} = 4$ variables using the SLSQP method.

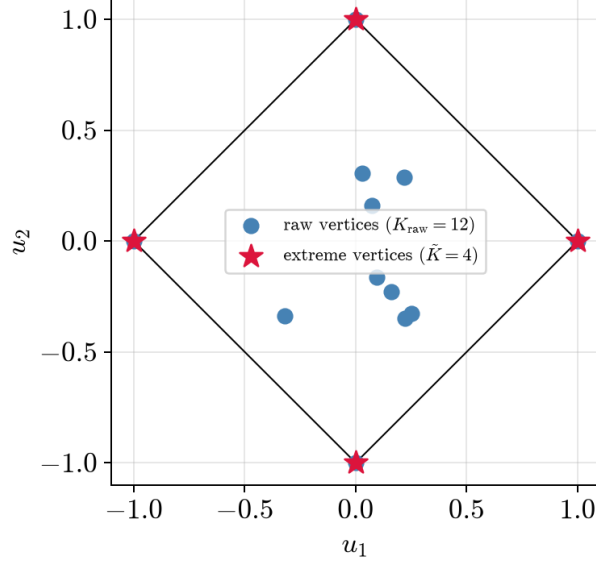


Figure 1: Raw vertex set U_{raw} ($K_{\text{raw}} = 12$, blue circles: the four true extreme vertices of the ℓ^1 diamond plus eight redundant interior points) and the four extreme vertices $\tilde{K} = 4$ recovered by Quickhull (red stars), together with the convex hull of U_{raw} .

Neural network and training. The trial solution takes the form

$$x_\theta(t) = x_0 + t N_\theta(t), \quad (6.6)$$

where $N_\theta : \mathbb{R} \rightarrow \mathbb{R}^2$ is a fully connected neural network with tanh activation, three hidden layers of width 64, and Glorot-normal initialization. The parametrization (6.6) enforces $x_\theta(0) = x_0$ exactly for every θ , so the DR-PINN loss reduces to the pure inclusion term

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \text{dist}^2(\dot{x}_\theta(t_i), F(x_\theta(t_i))), \quad (6.7)$$

evaluated at $N = 300$ collocation points distributed uniformly on $[0, T]$. Since $F(x_\theta(t_i)) = Ax_\theta(t_i) + B\tilde{U}$ is a translation of the *fixed* polytope $B\tilde{U}$, the residual is evaluated in the equivalent translated form $\text{dist}^2(\dot{x}_\theta(t_i) - Ax_\theta(t_i), B\tilde{U})$ of Remark 4.2(i): the projection is computed numerically by the QP (4.4) and the projected point is held constant in the backward pass, so automatic differentiation returns the exact gradients $2q$ and $-2A^\top q$ with respect to $\dot{x}_\theta$ and x_θ , respectively (the same practice as in the DR-PINN

companion run of Section 6.2). Training is performed with the Adam optimizer at learning rate 10^{-3} for 1000 epochs. Figure 2 shows the inclusion loss $\mathcal{L}(\theta)$ of (6.7) along training. Starting from $\mathcal{L}(\theta) = 9.6 \times 10^{-2}$ at initialization, the loss decreases to 4.1×10^{-14} after 1000 epochs, confirming that the learned velocity $\dot{x}_\theta(t)$ becomes admissible, i.e. $\dot{x}_\theta(t) \in F(x_\theta(t))$, at (numerically) every collocation point. Collocation-point accuracy does not control the residual between collocation points, so we also report an independent dense-grid validation below.

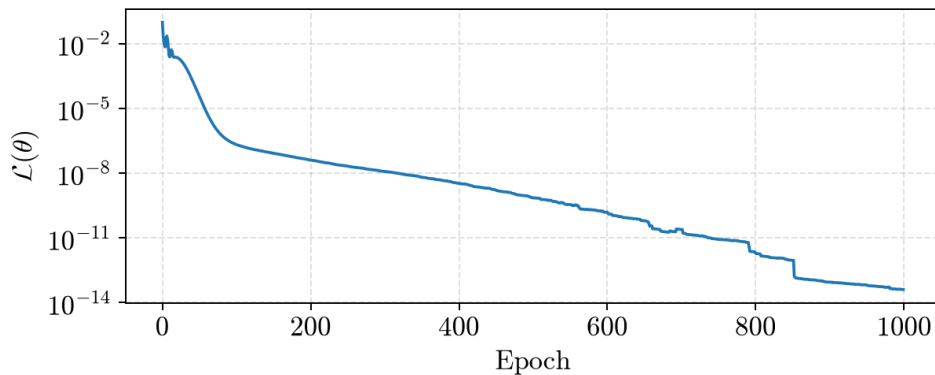


Figure 2: Training history of the inclusion loss $\mathcal{L}(\theta)$ for the single DR-PINN of this section (semi-logarithmic scale).

Reference reachable tube. A reference inner approximation of the reachable tube

$$\mathcal{R}(t) = \{x(t) : \dot{x}(s) = Ax(s) + Bu(s), u(s) \in U \text{ a.e.}, x(0) = x_0\} \quad (6.8)$$

is constructed by integrating the ODE $\dot{x} = Ax + Bu$ with two families of admissible controls:

- (a) *Constant bang-bang controls*: one trajectory per extreme vertex $u_k \in \tilde{U}$, totalling four reference trajectories. By Pontryagin's maximum principle, since the Hamiltonian of a linear system is affine in the control, boundary trajectories of $\mathcal{R}(t)$ are generated by bang-bang controls taking values in the extreme points $\tilde{U}$, possibly *switching* among them in time (see, e.g., [20], Chapter 4, or [2], Chapter 8). The four *constant* vertex controls therefore provide representative extremal trajectories; they do not, in general, trace the entire reachable-set boundary.

- (b) *Piecewise-constant random controls*: 300 trajectories obtained by randomly switching among the vertices of U at uniformly distributed times. Each constant-control segment $[t_j, t_{j+1}]$ is integrated over its *exact* interval with a fifth-order Runge–Kutta scheme (relative tolerance 10^{-9} , absolute tolerance 10^{-11}) and dense output; the state handed to the next segment is the state at exactly the switching time t_{j+1} , and the trajectory is then evaluated at the global grid times falling inside the segment. Since the system is linear with piecewise-constant control, the exact solution is available in closed form via the segment-wise matrix-exponential propagation $x(t_{j+1}) = e^{A\Delta t_j}x(t_j) + A^{-1}(e^{A\Delta t_j} - I)Bu_j$; the replication script cross-checks every integrated trajectory against this closed form and aborts if the deviation exceeds 10^{-7} (observed maximum over all trajectories and grid times: 4.7×10^{-10}). This family provides an empirical sample of the interior of $\mathcal{R}(t)$ (it does not densely cover it in any mathematical sense).

The union of all 304 trajectories evaluated on a uniform time grid of 200 points constitutes the *sampled reference cloud* used for comparison: at each grid time t ,

$$\widehat{\mathcal{R}}_{\text{ref}}(t) := \{x^{(j)}(t) : j = 1, \dots, 304\} \subset \mathcal{R}(t).$$

We use separate notation: $\mathcal{R}(t)$ denotes the exact reachable set of (6.8), while $\widehat{\mathcal{R}}_{\text{ref}}(t)$ is its finite empirical sample. Figure 3 overlays the trained trajectory $x_\theta(t)$ with the sampled reference cloud $\widehat{\mathcal{R}}_{\text{ref}}(t)$ approximating the reachable tube $\mathcal{R}(t)$ of (6.8): the left panel shows the phase portrait, the right panel the two state components against time. The learned trajectory remains within the visual envelope of the sampled reference trajectories throughout $[0, T]$, in line with the consistency mechanism of Theorem 4.8 (which concerns the continuous residual functional; cf. Remark 3.2): a small inclusion loss — corroborated here by the independent dense-grid residual validation reported below — produces a trajectory close to empirically generated admissible ones.

Ensemble and discrepancy from the sampled reference cloud. Since a single DR-PINN approximates one element of the solution set $\mathcal{S}(x_0)$ (cf. Section 4.3), its distance to the sampled reference cloud is not, by itself, a robust check. We therefore form an ensemble of $n_{\text{ens}} = 5$ trajectories (networks trained with different random seeds) and compute, at each time, the maximum distance of an ensemble member from the sampled reference cloud;

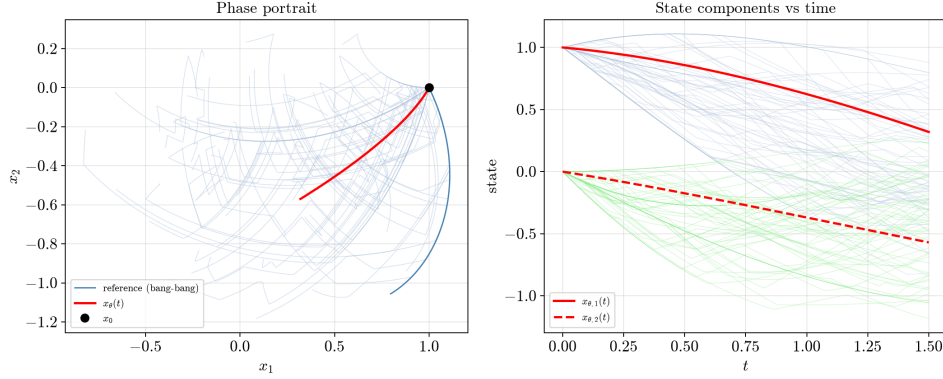


Figure 3: Phase portrait (left) and state components versus time (right) for the trained trajectory $x_\theta(t)$, compared with the sampled reference cloud $\hat{\mathcal{R}}_{\text{ref}}(t)$ (304 trajectories, light blue) and one representative bang-bang extremal (dark blue).

this quantity is then averaged over time. Precisely, the ensemble of trained trajectories at time t is

$$\mathcal{R}_\theta(t) = \{x_\theta^{(s)}(t) : s = 1, \dots, n_{\text{ens}}\},$$

and the discrepancy from the sampled reference cloud is measured by the one-sided Hausdorff distance

$$d_H^+(\mathcal{R}_\theta(t), \hat{\mathcal{R}}_{\text{ref}}(t)) = \max_{a \in \mathcal{R}_\theta(t)} \min_{b \in \hat{\mathcal{R}}_{\text{ref}}(t)} \|a - b\|,$$

which, at each time t , already takes the maximum over the ensemble trajectories; averaging this quantity over the uniform time grid gives the reported value

$$\overline{d_H^+}(\mathcal{R}_\theta, \hat{\mathcal{R}}_{\text{ref}}) = 0.0414,$$

which is small relative to the scale of the problem (the control set U and the reachable tube $\mathcal{R}(t)$ both have diameter of order 1). Because $\hat{\mathcal{R}}_{\text{ref}}(t)$ is only a finite trajectory sample, $\overline{d_H^+}$ measures one-sided discrepancy from that sample, not admissibility with respect to the exact reachable set $\mathcal{R}(t)$. Its value $\overline{d_H^+}$ also depends on the density of the cloud; a sparser cloud can increase the discrepancy even for admissible trajectories. We therefore use this quantity only to visualize the reachable tube. A certified comparison would require the sets $\mathcal{R}(t)$ themselves, for example through support functions or a guaranteed reachability solver. Admissibility in this experiment is assessed instead by the dense-grid residual validation reported next.

Independent dense-grid residual validation. As announced above, and analogously to the dense-grid check of Section 6.2, the inclusion residual of each ensemble member is additionally evaluated at 5 000 uniformly random times *off* the collocation grid, by solving the QP (4.4) for $\text{dist}(\dot{x}_\theta(t) - Ax_\theta(t), B\tilde{U})$ at each point. Over the whole ensemble, the worst-member mean inclusion distance is 3.2×10^{-7} and the worst pointwise maximum is 6.0×10^{-4} , confirming that the trained velocities remain (numerically) admissible between collocation points as well.

Computational scalability. Computational efficiency is assessed by a scalability study that varies $K_{\text{raw}} \in \{8, 16, 32, 64, 128\}$ (at fixed $d = 2$) and $d \in \{2, 4, 6, 8\}$ (at fixed $K_{\text{raw}} = 32$), comparing the per-epoch training time with and without the Quickhull reduction. Table 1 lists the number of extreme vertices $\tilde{K}$ recovered by Quickhull for each raw vertex-set size $K_{\text{raw}} \in \{8, 16, 32, 64, 128\}$, at fixed $d = 2$. As in the main example, the raw point cloud consists of a fixed hexagonal boundary (six true extreme points) plus $K_{\text{raw}} - 6$ redundant interior points, so $\tilde{K} = 6$ irrespective of K_{raw} .

K_{raw}	8	16	32	64	128
$\tilde{K}$	6	6	6	6	6

Table 1: Extreme vertices $\tilde{K}$ recovered by Quickhull as K_{raw} grows, at fixed $d = 2$.

For the dimension sweep, a fresh random drift matrix $A_d \in \mathbb{R}^{d \times d}$ is generated for each $d \in \{2, 4, 6, 8\}$, together with a random polytope of $K_{\text{raw}} = 32$ vertices. To guarantee that the uncontrolled system $\dot{x} = A_d x$ remains asymptotically stable regardless of the random draw, A_d is constructed by first sampling a Gaussian matrix G (i.i.d. entries $\mathcal{N}(0, 0.3^2)$) and then shifting its diagonal,

$$(A_d)_{ii} = G_{ii} - \left(\sum_{j=1}^d |G_{ij}| + 0.5 \right), \quad (A_d)_{ij} = G_{ij} \quad (j \neq i).$$

By the Gershgorin circle theorem, every eigenvalue λ of A_d lies in a disc centred at the real number $(A_d)_{ii}$ with radius $R_i = \sum_{j \neq i} |(A_d)_{ij}| = \sum_{j \neq i} |G_{ij}|$, so

$$\text{Re}(\lambda) \leq (A_d)_{ii} + R_i = G_{ii} - |G_{ii}| - 0.5 \leq -0.5$$

for every row i , since $G_{ii} \leq |G_{ii}|$. Hence every eigenvalue of A_d has real part at most -0.5 , for every random draw and every d : the construction guarantees a uniform stability margin without ever having to reject an unstable sample.

Table 2 shows that, unlike the K -sweep, $\tilde{K}$ now grows with d : a random point cloud in higher dimensions has a larger fraction of extreme points, and at $d = 6, 8$ all 32 raw vertices are already extreme, so no reduction is possible.

d	2	4	6	8
$\tilde{K}$	12	23	32	32

Table 2: Extreme vertices $\tilde{K}$ recovered by Quickhull as the state dimension d grows, at fixed $K_{\text{raw}} = 32$.

All timings reported here are collected in the same single execution of the replication script that produces every other number and figure of this section: for each configuration, one discarded warm-up training run is followed by 5 timed repetitions, and we report the mean per-epoch time (error bars in Figure 4 show one standard deviation across repetitions). The hardware and library versions of the run are recorded in the accompanying manifest of the replication package; absolute times are hardware-dependent, and only the *relative* comparison between the two variants is of interest.

Figure 4 reports the resulting per-epoch training time with and without the offline Quickhull reduction. As a function of K_{raw} (left panel), the Quickhull-reduced training time stays essentially flat, between 241 ms and 251 ms, while the unreduced QP grows from 258 ms at $K_{\text{raw}} = 8$ to 3.14 s at $K_{\text{raw}} = 128$ – a $12.5\times$ slowdown – because the QP of Section 4.2 then has 128 variables instead of 6. As a function of d (right panel), the benefit shrinks as $\tilde{K}$ approaches K_{raw} : at $d = 6$ and $d = 8$, where Table 2 shows no reduction is achieved, the Quickhull-reduced time (1.00 s, 0.91 s) is no longer lower than the unreduced time (1.13 s, 0.91 s). Since the Quickhull reduction is performed *offline* and does not contribute to the per-epoch time, when $\tilde{K} = K_{\text{raw}}$ the two variants solve identical QPs, and the residual differences between their timings are within run-to-run measurement variability rather than a preprocessing cost.

6.2. A Planar Differential Inclusion with a Rotating Ellipsoidal Control Constraint

Problem formulation. The two examples in Sections 6.1 and 6.3 enforce the inclusion by *minimizing* the distance residual. This second ODE example illustrates the complementary strategy anticipated in Section 4.1: when the admissible set has a tractable parametric description, admissibility can be built into the parametrization itself, so that the distance residual vanishes at

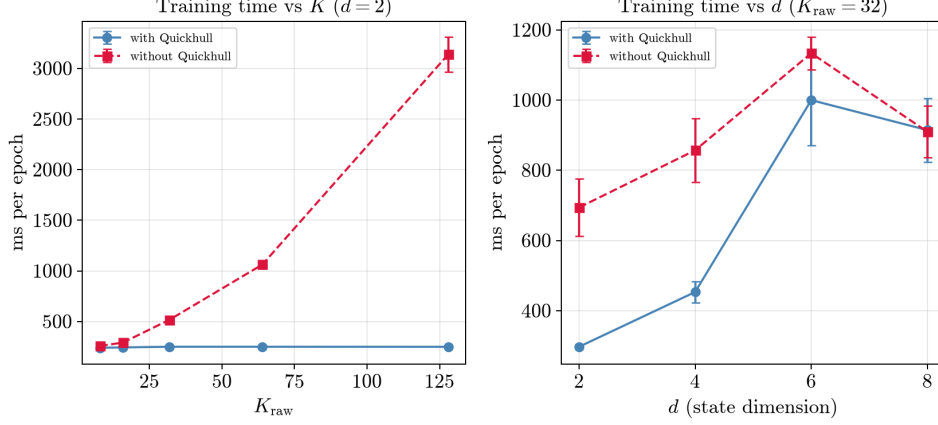


Figure 4: Per-epoch training time with (blue) and without (red) the offline Quickhull reduction; markers are means over 5 repetitions after a discarded warm-up run, error bars show one standard deviation. Left: as a function of K_{raw} at fixed $d=2$ (Table 1). Right: as a function of the state dimension d at fixed $K_{\text{raw}}=32$ (Table 2).

every time-grid node of every candidate trajectory and the remaining degrees of freedom can be used to select a specific element of the solution set $\mathcal{S}(x_0)$ — here, the trajectory steering the state closest to a prescribed terminal point. This also reflects the non-uniqueness discussed in Section 4.3: the distance-residual loss guarantees convergence to *some* admissible trajectory, whereas a hard-admissible parametrization allows one to optimize *which* one.

We consider trajectories $x : [0, T] \rightarrow \mathbb{R}^2$ of the differential inclusion

$$\dot{x}(t) \in F(t, x(t)) = g(t, x(t)) + E(t), \quad x(0) = x_0, \quad (6.9)$$

whose admissible-velocity set is of Minkowski type: a nonlinear, nonautonomous drift

$$g(t, x) = \begin{pmatrix} \sin(x_1) + 0.15 \cos(\frac{2\pi t}{T}) \\ 0.5 \cos(x_2) - 0.20 \sin(\frac{2\pi t}{T}) \end{pmatrix}, \quad x = (x_1, x_2)^\top, \quad (6.10)$$

translated by a compact, convex, *time-dependent* control set $E(t)$, a rotating ellipse defined below. Equivalently, admissible trajectories are exactly the solutions of the controlled system

$$\dot{x}(t) = g(t, x(t)) + \xi(t), \quad \xi(t) \in E(t) \text{ a.e. on } [0, T], \quad (6.11)$$

where ξ is a measurable selector of $t \mapsto E(t)$. Since $E(t)$ does not depend on the state, this example falls into regime (a) of Section 4.2; however, unlike

the polytopic set of Section 6.1, the projection onto an ellipse admits no closed form (it requires the solution of a scalar secular equation), so we also consider the hard-admissible route here. We take $T = 3$ and $x_0 = (0, 0)^\top$.

The rotating ellipsoidal control set. For $t \in [0, T]$, let

$$a(t) = 0.3 + 0.4 \left| \sin\left(\frac{\pi t}{T}\right) \right|, \quad b(t) = 0.15 + 0.25 \left| \cos\left(\frac{\pi t}{T}\right) \right|, \quad \varphi(t) = \frac{\pi t}{T}, \quad (6.12)$$

and let $R_{\varphi(t)}$ denote the rotation by the angle $\varphi(t)$. The control set is the (filled) rotating ellipse

$$E(t) = \left\{ R_{\varphi(t)} \begin{pmatrix} u \\ w \end{pmatrix} \in \mathbb{R}^2 : \frac{u^2}{a(t)^2} + \frac{w^2}{b(t)^2} \leq 1 \right\}, \quad (6.13)$$

which is nonempty, compact, and convex for every t , with semi-axes and orientation varying continuously in time; since $\varphi(0) = 0$ and $\varphi(T) = \pi$, the ellipse performs a half-turn over the horizon, and the translated family $g(t, x(t)) + E(t)$ traces a twisted tube of admissible velocities along the drift curve (Figure 5). Writing $(u, w)^\top = R_{-\varphi(t)} \xi$ for the coordinates of $\xi \in \mathbb{R}^2$ in the local frame of the ellipse, the *level function*

$$\ell(t, \xi) = \left(\frac{u}{a(t)} \right)^2 + \left(\frac{w}{b(t)} \right)^2 \quad (6.14)$$

characterizes admissibility: $\xi \in E(t)$ if and only if $\ell(t, \xi) \leq 1$, with $\ell = 1$ on the boundary $\partial E(t)$.

Hard-admissible selector parametrization. The selector is generated using $n_c = 24$ control nodes $(p^j, \rho^j) \in \mathbb{R}^2 \times \mathbb{R}$, $j = 0, \dots, n_c - 1$, linearly interpolated to the uniform time grid $t_i = i \Delta t$, $i = 0, \dots, n_t - 1$, $\Delta t = T/(n_t - 1)$, with $n_t = 120$; the low-dimensional nodal representation suppresses artificial oscillations of the selector. From the interpolated values $p(t_i) \in \mathbb{R}^2$, $\rho(t_i) \in \mathbb{R}$, define the unit-ball direction and the sigmoidal radius

$$d(t_i) = \frac{p(t_i)}{\sqrt{|p(t_i)|^2 + \varepsilon}}, \quad s(t_i) = \frac{1}{1 + e^{-\rho(t_i)}} \in (0, 1), \quad (6.15)$$

with $\varepsilon = 10^{-10}$ a normalization safeguard, and set

$$\xi(t_i) = R_{\varphi(t_i)} \begin{pmatrix} s(t_i) a(t_i) d_1(t_i) \\ s(t_i) b(t_i) d_2(t_i) \end{pmatrix}. \quad (6.16)$$

Substituting (6.16) into (6.14) gives, for every grid point,

$$\ell(t_i, \xi(t_i)) = s(t_i)^2 |d(t_i)|^2 \leq s(t_i)^2 < 1, \quad (6.17)$$

so the selector lies in the interior of $E(t_i)$ *by construction*, for every value of the nodal variables: no penalty for constraint violation is required, and the inclusion residual $\text{dist}(\dot{x}(t_i) - g(t_i, x(t_i)), E(t_i))$ of Section 4.1 vanishes at every grid point t_i . The guarantee (6.17) is nodewise: ξ is recomputed from (6.16) at the grid points. Between nodes, the selector ξ is linearly interpolated from the values $\xi(t_i)$, and a convex combination of velocities admissible at different times need not lie in the intermediate rotated and rescaled ellipse. For the reported optimized selector we verified admissibility a posteriori on a dense grid of 20 001 intermediate points: the maximum level is $\ell \approx 0.987$, with no violations. The dense-grid maximum may therefore slightly exceed the grid-node maximum in Table 3: the former also probes the intermediate ellipses, which are not covered by the nodewise construction (6.17). An implementation that re-evaluates $d(t)$, $s(t)$, and $\xi(t)$ from interpolated nodal values $p(t)$, $\rho(t)$ at every RK4 stage would make the guarantee hold at every evaluation time by construction. This is the analogue, for the inclusion constraint itself, of the hard-constraint ansatz used for the initial condition in Section 6.1 and for the initial/boundary conditions in Section 6.3.

Steering problem and discretization. For a fixed selector, the state is propagated through (6.11) by the classical fourth-order Runge–Kutta scheme on the grid $\{t_i\}$, with ξ evaluated between grid points by linear interpolation. The nodal variables $z = (p^0, \dots, p^{n_c-1}, \rho^0, \dots, \rho^{n_c-1}) \in \mathbb{R}^{3n_c} = \mathbb{R}^{72}$ are chosen by minimizing

$$J(z) = \lambda_f |x_z(T) - x_f|^2 + \lambda_\xi J_\xi(z) + \lambda_s J_s(z) + \lambda_c J_c(z) + \lambda_b J_b(z) + \lambda_u J_u(z), \quad (6.18)$$

where x_z denotes the discrete trajectory induced by z , the terminal term steers the state to a prescribed target x_f , and the regularizers are the mean level $J_\xi = \frac{1}{n_t} \sum_i \ell(t_i, \xi_z(t_i))$ (discouraging boundary-hugging selectors), the mean squared first and second discrete differences of ξ_z (J_s , J_c : smoothness and curvature), the cubic soft barrier $J_b = \frac{1}{n_t} \sum_i \ell(t_i, \xi_z(t_i))^3$ (penalizing boundary saturation), and the mean squared differences of the nodal variables (J_u). The weights are $\lambda_f = 2500$, $\lambda_s = 1$, $\lambda_c = 0.2$, $\lambda_b = 10^{-2}$, $\lambda_u = 5 \cdot 10^{-2}$, and the selector-regularization weight is driven along the continuation sequence $\lambda_\xi \in \{10^{-3}, 10^{-2}, 10^{-1}\}$, each stage warm-started from the previous one and solved by L-BFGS-B.

The target must of course be chosen inside the reachable set of (6.9): the drift component $g_2 = 0.5 \cos(x_2) - 0.2 \sin(2\pi t/T)$ pushes the state upward with magnitude ≈ 0.5 near $x_2 = 0$, while the vertical half-width of the rotated ellipse $E(t)$,

$$h_y(t) = \sqrt{a(t)^2 \sin^2 \varphi(t) + b(t)^2 \cos^2 \varphi(t)},$$

stays between ≈ 0.391 and 0.7 for the parameters (6.12) (the value 0.15 is the minimal length of one semi-axis, not the minimal vertical reach of the rotated ellipse), so targets with markedly negative x_2 are unreachable over $[0, T]$ (an extremal selector taking, at every instant, the support point of $E(t)$ in the direction $-e_2$ only attains $x_2(T) \approx -0.05$). We take $x_f = (1.8, 0.1)^\top$, which lies inside but close to the boundary of the reachable set: steering to it forces the selector to operate near $\partial E(t)$ over most of the horizon, so the built-in admissibility guarantee (6.17) is genuinely stressed.

Numerical results. Table 3 reports, for each continuation stage, the terminal mismatch and the mean and maximum of the level function along the optimized selector. The terminal state matches the target closely at every stage, while increasing λ_ξ pulls the selector towards the interior of the ellipse (mean level decreasing from 0.972 to 0.968) at essentially no cost in terminal accuracy. For a target close to the reachable-set boundary, near-extremal velocities are needed over most of the horizon.

λ_ξ	$ x_z(T) - x_f $	mean ℓ	max ℓ
10^{-3}	2.0×10^{-5}	0.972	0.982
10^{-2}	4.8×10^{-6}	0.971	0.982
10^{-1}	9.7×10^{-6}	0.968	0.982

Table 3: Continuation over the selector-regularization weight λ_ξ (warm-started L-BFGS-B): terminal mismatch and level statistics of the optimized selector. Admissibility $\ell(t_i, \xi_z(t_i)) < 1$ holds at every grid point and every stage, by construction.

Figure 5 displays the resulting geometry in the space (v_1, v_2, t) of velocities and time: the translated ellipses $g(t_i, x_z(t_i)) + E(t_i)$ form a twisted tube whose centerline is the drift curve, and the selected velocity $\dot{x}_z(t_i) = g(t_i, x_z(t_i)) + \xi_z(t_i)$ runs inside the tube, close to its wall, as quantified by the level statistics of Table 3. Figure 6 shows the same information in selector coordinates: the trajectory $t \mapsto \xi_z(t)$ threads the rotating ellipse $E(t)$ centered at the origin, with the color scale reporting the level $\ell(t_i, \xi_z(t_i))$.

Finally, Figure 7 shows the optimized state trajectory from x_0 to x_f (left) together with the admissibility level along the horizon (right), which stays below the boundary value $\ell = 1$ on the dense validation grid.

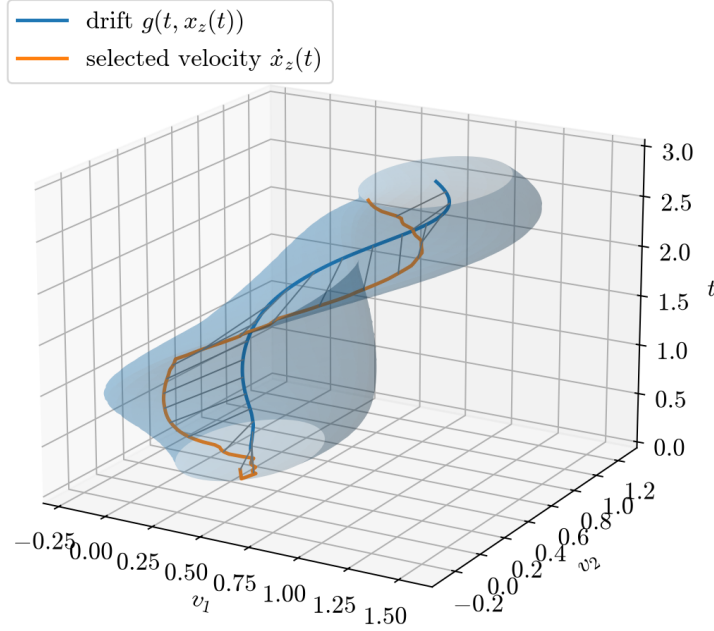


Figure 5: Admissible velocity tube of the inclusion (6.9) in (v_1, v_2, t) space. The surface is the boundary of the translated rotating ellipses $g(t, x_z(t)) + E(t)$; the blue curve is the centerline $g(t, x_z(t))$, the orange curve the selected velocity $\dot{x}_z(t)$, and the grey segments the selector displacements $\xi_z(t_i)$.

DR-PINN companion run. The hard-admissible selector above is a constrained optimal-control discretization: it propagates the state by RK4 and enforces the inclusion at the time-grid nodes by construction. We therefore use a separate DR-PINN run to test the distance-residual method on the same benchmark. Because the experiment is a steering problem with prescribed initial and terminal states, we used the endpoint-hard neural ansatz

$$x_\theta(t) = x_0 + \frac{t}{T}(x_f - x_0) + t(T - t)N_\theta(t), \quad (6.19)$$

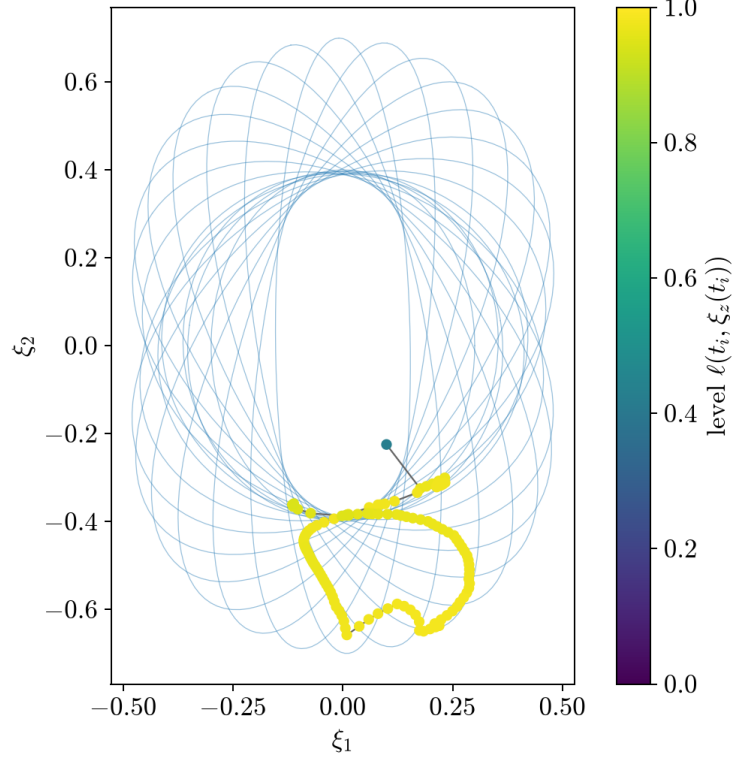


Figure 6: Optimized selector $\xi_z(t)$ inside the rotating control ellipse $E(t)$ (thin blue rings, drawn every sixth grid point). The color of the points encodes the level $\ell(t_i, \xi_z(t_i))$, which remains below 1 at the plotted time-grid nodes by construction.

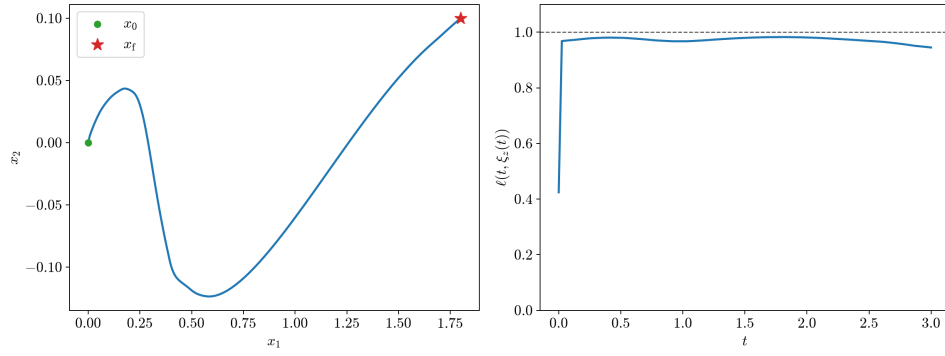


Figure 7: Left: optimized state trajectory of (6.11) from $x_0 = (0,0)^\top$ to the target $x_f = (1.8, 0.1)^\top$ (final continuation stage $\lambda_\xi = 10^{-1}$, terminal mismatch 9.7×10^{-6}). Right: admissibility level $\ell(t, \xi_z(t))$ along the horizon; the dashed line marks the boundary value $\ell = 1$ of $\partial E(t)$.

where N_θ is a 3×64 tanh network. Hence $x_\theta(0) = x_0$ and $x_\theta(T) = x_f$ exactly, and the training objective is the pure distance residual

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \text{dist}^2(\dot{x}_\theta(t_i) - g(t_i, x_\theta(t_i)), E(t_i)), \quad (6.20)$$

with $N = 512$ uniform collocation points.

The projection onto an ellipse is evaluated pointwise by Newton iteration on the secular equation $\sum_k (\sigma_k v_k^{\text{loc}} / (\sigma_k^2 + \mu))^2 = 1$ in the local frame of $E(t)$, with $\sigma_1 = a(t)$ and $\sigma_2 = b(t)$. In the backward pass the *entire projected point* $\Pi_{E(t)}(v)$ is held constant, so automatic differentiation returns the exact envelope gradient $2(v - \Pi_{E(t)}(v))$. Holding only the multiplier μ constant would not give this gradient; this distinction is important in an implementation with a numerically computed projection. A finite-difference gradient check in the replication code gives a relative error of 7.2×10^{-11} .

After 7000 Adam epochs (hold-then-decay learning rate, double precision), the collocation loss is 6.9×10^{-19} . On an independent dense grid of 2000 time points, the mean and maximum inclusion distances are respectively 2.1×10^{-9} and 1.4×10^{-6} , while both endpoint errors are zero up to machine precision. The corresponding selector has mean level 0.923 and maximum level 1.000006; any excess of the latter over 1 is consistent with the reported maximum inclusion distance and the numerical projection tolerance. With this implementation, the DR-PINN attains the reported accuracy on this benchmark.

The companion result is based on a single random seed. It establishes the reported accuracy for this run but does not characterize training robustness. A fuller robustness study would report medians and ranges over 5–10 seeds, unsuccessful-run frequency, and a comparison with a direct parametrization of the control ξ trained on the same steering objective. The hard-admissible selector is a complementary constrained discretization, not a DR-PINN experiment. Conclusions about the distance residual rely on the companion run and on the experiments of Sections 6.1 and 6.3. The hard-admissible parametrization guarantees feasibility at the time-grid nodes independently of training; feasibility between nodes for the reported selector was also checked on a dense grid of 20 001 points.

6.3. A Reaction-Diffusion Inclusion with Relay (Sliding-Mode) Feedback

As a benchmark for the parabolic setting of Section 5, we consider the reaction-diffusion inclusion

$$\partial_t u(t, x) - \Delta u(t, x) \in \Phi(u(t, x)), \quad (t, x) \in Q = (0, T) \times \Omega, \quad (6.21)$$

with homogeneous Dirichlet boundary condition, where the set-valued reaction term is the relay (signum) multifunction

$$\Phi(s) := -\lambda \operatorname{Sign}(s) = \begin{cases} \{-\lambda\}, & s > 0, \\ [-\lambda, \lambda], & s = 0, \\ \{\lambda\}, & s < 0, \end{cases} \quad \lambda > 0. \quad (6.22)$$

Physically, (6.21)–(6.22) are the parabolic analogue of the bang-bang differential inclusion of Section 6.1. Indeed, at every point (t, x) , instead of a smooth reaction term, the state is driven towards the equilibrium $u \equiv 0$ by a switching (relay) feedback of constant magnitude λ , exactly as a sliding-mode controller drives a state to a manifold at constant rate regardless of its distance to that manifold. Equivalently, $\Phi = -\lambda \partial|\cdot|$, so that (6.21) may be read as the abstract evolution inclusion

$$\partial_t u \in -\partial E(u), \quad E(u) = \frac{1}{2} \|\nabla u\|_{L^2(\Omega)}^2 + \lambda \|u\|_{L^1(\Omega)}, \quad (6.23)$$

associated with the Dirichlet energy regularized by an L^1 (total-mass) penalty, which connects this example to sparsity-promoting flows in variational image processing and signal denoising.

We take $\Omega = (0, 1)^2$ and

$$u_0(x, y) = 16 x(1 - x) y(1 - y), \quad (6.24)$$

which is smooth, nonnegative, and satisfies $u_0 \in H_0^1(\Omega)$, and take $\lambda = 0.6$ as a baseline value, together with a primary comparison sweep $\lambda \in \{0.4, 0.6, 0.8\}$ to illustrate how the extinction time decreases as λ increases (cf. Proposition 6.2 below). The time horizon is fixed at $T = 0.3$, which is large enough that all three relay cases have visibly reached extinction while the $\lambda = 0$ (classical heat equation) case has not, making the qualitative contrast visible in a single plot (see Figure 8 below).

We check that Φ in (6.22) satisfies (P1)–(P3) of Section 2.4, so that both the existence result (Theorem 2.20) and the consistency result (Theorem 5.3) apply to this example.

- (P1) For every $s \in \mathbb{R}$, $\Phi(s)$ is a singleton or the closed interval $[-\lambda, \lambda]$, hence nonempty, closed, and convex.
- (P2) Φ is upper semicontinuous: it is locally constant on $\mathbb{R} \setminus \{0\}$, and at $s = 0$, $\Phi(s_n) \subset [-\lambda, \lambda] = \Phi(0)$ for every sequence $s_n \rightarrow 0$. This is the standard Filippov regularization of the signum function [13].

(P3) $\sup_{r \in \Phi(s)} |r| = \lambda \leq \lambda(1 + |s|)$ for every $s \in \mathbb{R}$, so (P3) holds with $m_0 = \lambda$.

Remark 6.1. Existence of a strong solution for this example is guaranteed directly by Theorem 2.20, since (P1)–(P3) hold. Note that any selection of Φ in (6.22) is forced to equal $-\lambda$ for $s > 0$ and λ for $s < 0$, hence is necessarily discontinuous at $s = 0$; no reduction to a single-valued semilinear equation via a continuous selection is available, and the genuinely multivalued fixed-point argument of Section 2.4 is essential here. The example has, however, additional structure that yields more than existence — namely uniqueness and explicit a priori bounds — via the classical theory of evolution equations governed by maximal monotone operators [8]: rewriting (6.21) as $\partial_t u + (-\Delta + \lambda \partial|\cdot|)(u) \ni 0$, we identify $-\Delta + \lambda \partial|\cdot|$ (with the Dirichlet Laplacian) as the subdifferential ∂E of the functional E of (6.23), extended by $E(u) = +\infty$ for $u \in L^2(\Omega) \setminus H_0^1(\Omega)$. Indeed, $E = \varphi_1 + \varphi_2$ with $\varphi_1(u) = \frac{1}{2} \|\nabla u\|_{L^2(\Omega)}^2$ proper, convex, and lower semicontinuous on $L^2(\Omega)$, and $\varphi_2(u) = \lambda \|u\|_{L^1(\Omega)}$ convex and, since Ω is bounded, finite and continuous on all of $L^2(\Omega)$; the subdifferential sum rule (Moreau–Rockafellar, with the qualification condition satisfied because φ_2 is continuous) then gives $\partial E = \partial \varphi_1 + \partial \varphi_2 = -\Delta + \lambda \partial|\cdot|$. As the subdifferential of a proper, convex, lower semicontinuous functional, ∂E is maximal monotone [8]. Here maximal monotonicity follows from the subdifferential representation; maximal monotonicity of the two summands alone would not suffice in general.

The subgradient-flow theory provides existence in L^2 and also yields the regularity required for a strong solution in the sense of Definition 2.11, defined for every time horizon. Indeed, since $u_0 \in H_0^1(\Omega)$ and Ω is bounded, $E(u_0) < \infty$, and the theory of subgradient flows with initial datum in the domain of the energy [8] produces, for every $T > 0$, a unique function $u \in H^1(0, T; L^2(\Omega))$ with $u(t) \in D(\partial E)$ and $-\partial_t u(t) \in \partial E(u(t))$ for a.e. t . By the subdifferential sum rule above, for a.e. t one has $u(t) \in H^2(\Omega) \cap H_0^1(\Omega)$ and there exists $\xi(t, \cdot) \in \partial \|\cdot\|_{L^1(\Omega)}(u(t))$, i.e., $\xi(t, x) \in \text{Sign}(u(t, x))$ a.e., such that

$$\partial_t u - \Delta u + \lambda \xi = 0 \quad \text{a.e. on } Q.$$

The selection is automatically measurable, since $\xi = (\Delta u - \partial_t u)/\lambda$ pointwise a.e., and $|\xi| \leq 1$ gives $\xi \in L^\infty(Q) \subset L^2(Q)$. Consequently,

$$\Delta u = \partial_t u + \lambda \xi \in L^2(Q),$$

and the H^2 elliptic regularity on convex or $C^{1,1}$ domains (see [14, Theorem 3.2.1.2] and recall that $\Omega = (0, 1)^2$ is convex) yields $u \in L^2(0, T; H^2(\Omega) \cap H_0^1(\Omega))$.

$H_0^1(\Omega)$). Hence $u \in \mathcal{W}(0, T)$ is a strong solution of (6.21), recovering $\mathcal{S}(u_0) \neq \emptyset$ independently of Theorem 2.20.

The stated regularity can also be read off from explicit L^2 estimates. For gradient flows $-\partial_t u(t) \in \partial E(u(t))$ with $u_0 \in D(E)$, the chain rule for convex subdifferentials [8, Lemme 3.3] gives $\frac{d}{dt} E(u(t)) = -\|\partial_t u(t)\|_{L^2(\Omega)}^2$ for a.e. t , whence, integrating on $(0, T)$ and using $E \geq 0$,

$$\begin{aligned} \int_0^T \|\partial_t u(t)\|_{L^2(\Omega)}^2 dt &= E(u_0) - E(u(T)) \leq E(u_0) \\ &= \frac{1}{2} \|\nabla u_0\|_{L^2(\Omega)}^2 + \lambda \|u_0\|_{L^1(\Omega)}. \end{aligned}$$

The pointwise bound $|\xi| \leq 1$ a.e. on Q also gives

$$\|\xi\|_{L^2(Q)} \leq |Q|^{1/2} = (T |\Omega|)^{1/2}.$$

Combining these bounds with the triangle inequality in $L^2(Q)$,

$$\|\Delta u\|_{L^2(Q)} \leq \|\partial_t u\|_{L^2(Q)} + \lambda \|\xi\|_{L^2(Q)} \leq E(u_0)^{1/2} + \lambda (T |\Omega|)^{1/2},$$

and the constant in the elliptic estimate $\|u(t)\|_{H^2(\Omega)} \leq C(\Omega) \|\Delta u(t)\|_{L^2(\Omega)}$ of [14, Theorem 3.2.1.2] depends only on the (convex) domain. All three memberships $\partial_t u \in L^2(0, T; L^2(\Omega))$, $\xi \in L^2(Q)$, and $u \in L^2(0, T; H^2(\Omega) \cap H_0^1(\Omega))$ therefore come with explicit bounds in terms of $E(u_0)$, λ , T , and $|\Omega|$ alone.

Uniqueness of strong solutions also holds. Let $u_1, u_2 \in \mathcal{W}(0, T)$ be strong solutions with the same initial datum, with pointwise selections $\xi_i(t, x) \in \text{Sign}(u_i(t, x))$ such that $\partial_t u_i - \Delta u_i = -\lambda \xi_i$ a.e. on Q . Subtracting the two equations and testing with $u_1 - u_2$ gives, for a.e. $t \in (0, T)$,

$$\frac{1}{2} \frac{d}{dt} \|u_1 - u_2\|_{L^2(\Omega)}^2 + \|\nabla(u_1 - u_2)\|_{L^2(\Omega)}^2 + \lambda \int_{\Omega} (\xi_1 - \xi_2)(u_1 - u_2) dx = 0.$$

Since the graph of Sign is monotone, $(\xi_1 - \xi_2)(u_1 - u_2) \geq 0$ a.e., whence $\frac{d}{dt} \|u_1 - u_2\|_{L^2(\Omega)}^2 \leq 0$; as $u_1(0) = u_2(0) = u_0$, we conclude $u_1 = u_2$. Thus $\mathcal{S}(u_0)$ is a singleton, in sharp contrast with the genuinely multivalued solution sets of Section 6.1.

For the finite-time extinction analysis, multiply (6.21) by $u(t, \cdot)$ and integrate over Ω . Using that $s \cdot v = -\lambda |s|$ for every $v \in \Phi(s)$ and every $s \in \mathbb{R}$, a formal computation (rigorous for strong solutions) gives the energy identity

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_{L^2(\Omega)}^2 + \|\nabla u(t)\|_{L^2(\Omega)}^2 = -\lambda \|u(t)\|_{L^1(\Omega)}. \quad (6.25)$$

The extra dissipation term $-\lambda\|u(t)\|_{L^1(\Omega)}$, absent in the classical heat equation ($\lambda = 0$), is responsible for a qualitatively different phenomenon: the solution reaches the equilibrium $u \equiv 0$ in *finite time* rather than merely decaying exponentially — a behavior familiar from evolution problems governed by dissipation terms that are positively homogeneous of degree one, such as the total variation flow [1]. The energy identity (6.25) alone does not yield finite-time extinction. For the nonnegative initial datum (6.24), extinction, an explicit upper bound for the extinction time, and monotonicity with respect to λ follow from the comparison argument in the next proposition. Once $u(t, \cdot)$ vanishes on part of Ω , the relay may “slide” there ($\Delta u(t, x) \in [-\lambda, \lambda]$ pointwise, consistently with $u \equiv 0$), anchoring the solution at equilibrium. A classical pointwise-residual PINN would have to choose one branch of Φ in advance and would not represent this sliding behavior directly.

Proposition 6.2 (Finite-time extinction and monotonicity in λ). *Let $u_0 \in H_0^1(\Omega) \cap L^\infty(\Omega)$ with $u_0 \geq 0$, and, for $\lambda > 0$, let u_λ denote the unique strong solution of (6.21)–(6.22) provided by Remark 6.1, regarded as defined for all $t \geq 0$. Then:*

- (a) $u_\lambda(t, x) \geq 0$ for a.e. (t, x) ;
- (b) $0 \leq u_\lambda(t, x) \leq (\|u_0\|_{L^\infty(\Omega)} - \lambda t)_+$ a.e.; in particular, $u_\lambda(t, \cdot) \equiv 0$ for every $t \geq \|u_0\|_{L^\infty(\Omega)}/\lambda$, so the extinction time $t^*(\lambda) := \inf\{t \geq 0 : u_\lambda(t, \cdot) \equiv 0\}$ satisfies

$$t^*(\lambda) \leq \frac{\|u_0\|_{L^\infty(\Omega)}}{\lambda};$$

- (c) if $\lambda_2 > \lambda_1 > 0$, then $0 \leq u_{\lambda_2} \leq u_{\lambda_1}$ a.e., and consequently $t^*(\lambda_2) \leq t^*(\lambda_1)$.

Proof. Throughout, $u = u_\lambda$ represents the strong solution and $\xi(t, x) \in \text{Sign}(u(t, x))$ the associated selection with $\partial_t u - \Delta u = -\lambda \xi$ a.e. (cf. Remark 6.1); all the test computations below are rigorous for strong solutions.

(a) Let $u^- := \max(-u, 0)$, so that $u^-(t, \cdot) \in H_0^1(\Omega)$ for a.e. t . Multiplying the equation by $-u^-$ and integrating over Ω gives, for a.e. t ,

$$\frac{1}{2} \frac{d}{dt} \|u^-(t)\|_{L^2(\Omega)}^2 + \|\nabla u^-(t)\|_{L^2(\Omega)}^2 = \lambda \int_{\Omega} \xi u^- dx = -\lambda \int_{\Omega} u^- dx \leq 0,$$

where we used that $\xi = -1$ a.e. on $\{u < 0\}$ and $u^- = 0$ elsewhere. Since $u^-(0) = u_0^- = 0$, we conclude $u^- \equiv 0$, i.e., $u \geq 0$.

(b) Set $M := \|u_0\|_{L^\infty(\Omega)}$, $t_0 := M/\lambda$, and consider the spatially constant supersolution $v(t) := M - \lambda t$ on $[0, t_0]$. Let $w := u - v$. For $t \in (0, t_0)$ one has $v(t) > 0$, hence $w < 0$ on $(0, t_0) \times \partial\Omega$ and $w^+ := \max(w, 0)$ satisfies $w^+(t, \cdot) \in H_0^1(\Omega)$. Moreover, $\partial_t v - \Delta v = -\lambda$, so

$$\partial_t w - \Delta w = \lambda(1 - \xi) \quad \text{a.e. in } (0, t_0) \times \Omega,$$

and on $\{w > 0\}$ we have $u > v > 0$, hence $\xi = 1$ and the right-hand side vanishes there. Testing with w^+ therefore gives

$$\frac{1}{2} \frac{d}{dt} \|w^+(t)\|_{L^2(\Omega)}^2 + \|\nabla w^+(t)\|_{L^2(\Omega)}^2 = \lambda \int_{\{w>0\}} (1 - \xi) w^+ dx = 0,$$

and since $w^+(0) = (u_0 - M)^+ = 0$, we obtain $u \leq v$ on $[0, t_0]$. Together with (a), this proves the pointwise bound in (b) on $[0, t_0]$; by continuity of $t \mapsto u(t)$ in $L^2(\Omega)$, $u(t_0) = 0$. Finally, the identically zero function is a strong solution of (6.21) with zero initial datum (indeed $0 = \partial_t 0 - \Delta 0 \in [-\lambda, \lambda] = \Phi(0)$), so by the uniqueness established in Remark 6.1, restarted at time t_0 , $u(t) \equiv 0$ for all $t \geq t_0$.

(c) Let $w := u_{\lambda_2} - u_{\lambda_1}$, with selections $\xi_i \in \text{Sign}(u_{\lambda_i})$. Then $\partial_t w - \Delta w = -\lambda_2 \xi_2 + \lambda_1 \xi_1$, and on $\{w > 0\}$ we have $u_{\lambda_2} > u_{\lambda_1} \geq 0$ by (a), hence $\xi_2 = 1$ and $-\lambda_2 \xi_2 + \lambda_1 \xi_1 \leq -\lambda_2 + \lambda_1 < 0$. Both u_{λ_i} vanish on $\partial\Omega$, so $w^+(t, \cdot) \in H_0^1(\Omega)$, and testing with w^+ gives

$$\frac{1}{2} \frac{d}{dt} \|w^+(t)\|_{L^2(\Omega)}^2 + \|\nabla w^+(t)\|_{L^2(\Omega)}^2 \leq \int_{\{w>0\}} (\lambda_1 \xi_1 - \lambda_2) w^+ dx \leq 0.$$

With $w^+(0) = 0$ this yields $w^+ \equiv 0$, i.e., $u_{\lambda_2} \leq u_{\lambda_1}$, and the ordering of the extinction times follows immediately from the definition of t^* . $\square$

For the initial datum (6.24) one has $\|u_0\|_{L^\infty(\Omega)} = 1$, so Proposition 6.2(b) gives the a priori bound $t^*(\lambda) \leq 1/\lambda$, i.e., 2.5, 1.66, and 1.25 for $\lambda \in \{0.4, 0.6, 0.8\}$. The bound is far from sharp (the observed extinction times reported below are an order of magnitude smaller), since the comparison argument discards the additional decay produced by diffusion; its role is purely qualitative: it certifies finite-time extinction and the monotone dependence on λ . Identity (6.25) and Proposition 6.2 also provide solver-independent numerical checks: the decay of $\|u_\theta(t)\|_{L^2(\Omega)}^2$ along the trained trajectory should be strictly faster than that of the corresponding heat equation ($\lambda = 0$) with the same initial datum, and the detected extinction times must be nonincreasing in λ (strictly decreasing in our experiments).

Adaptive collocation sampling. Because Φ has a genuine jump discontinuity at $s = 0$, the distance residual is hardest to drive to zero in a neighborhood of the (a priori unknown, time-dependent) extinction front

$$\{(t, x) : u_\theta(t, x) = 0\}.$$

This is precisely where a smooth network output must approximate a discontinuous right-hand side, and our experiments confirm that the residual after training is concentrated there rather than spread uniformly over Q . Purely uniform collocation under-resolves this thin, moving region relative to its share of the loss. We therefore complement uniform collocation with a *front-focused, state-based* adaptive sampling scheme inspired by the RAR method of Lu, Meng, Mao, and Karniadakis [22]; unlike residual-based RAR, our scheme selects points by the smallest $|u_\theta|$ (a proxy for proximity to the extinction front) rather than by the largest residual. Precisely, at fixed intervals during training, a large pool of candidate points is drawn uniformly over Q , the points with the smallest $|u_\theta|$ are retained as an auxiliary “hard pool” tracking the current estimate of the extinction front, and a fraction of every subsequent training batch is drawn from this pool — refreshed periodically as the front moves — while the remaining points continue to be sampled uniformly over Q to preserve global coverage of the domain. In our experiments the batch size is $N_{\text{col}} = 2000$ collocation points per epoch; the adaptive fraction is 30% (600 points), drawn from a hard pool of $n_{\text{pool}} = 4000$ candidate points selected from a uniform draw of 20 000 points over Q by retaining those with smallest $|u_\theta|$. The hard pool is refreshed every 500 training epochs as the extinction front advances; the remaining 70% of each batch is drawn uniformly over Q to preserve global domain coverage. The campaign therefore includes a controlled sampler ablation (front-based adaptive vs. residual-based RAR vs. uniform collocation) at the baseline configuration. At these settings the three samplers are statistically indistinguishable on every reported metric (Table 7); the front-based scheme is retained as a safeguard for thinner bands rather than as a demonstrated speedup.

Reference solution. Unlike the bang-bang example of Section 6.1, where F is genuinely multivalued and the ensemble/Hausdorff comparison of Section 6.1 is required, uniqueness of strong solutions holds here, as proved in Remark 6.1: $\mathcal{S}(u_0)$ is a singleton. A single trained DR-PINN can therefore be validated directly against one reference solution, obtained by a first-order operator-splitting scheme on a uniform 49×49 interior grid with time step $\Delta t = 5 \times 10^{-4}$. Each time step alternates an implicit diffusion solve,

$v = (I - \Delta t \Delta_h)^{-1} u^n$ (with the 5-point finite-difference Laplacian Δ_h and homogeneous Dirichlet boundary conditions), and an exact proximal step,

$$u^{n+1} = \text{Sign}(v) \max(|v| - \lambda \Delta t, 0),$$

the pointwise proximal operator of $s \mapsto \lambda \Delta t |s|$. This scheme is able to represent exact zeros: any grid point for which $|v_i| \leq \lambda \Delta t$ is set to exactly zero *at that time step* by the soft-thresholding operation (the subsequent diffusion half-step may of course make it nonzero again). In particular, once the whole field has decayed below the threshold, the scheme reproduces the identically zero equilibrium without smoothing, so finite-time extinction is captured sharply. Being a first-order splitting on a fixed grid, the scheme carries discretization errors of order $O(\Delta t) + O(h^2)$ that may systematically shift the detected extinction instants; a refinement study therefore quantifies the reliability of the reported t_{ref}^* . Table 4 reports, for each solver configuration, the detected extinction time t^* (first grid time at which the discrete field is exactly zero; the zero state is persistent up to T in every run), its change Δt^* with respect to the baseline configuration, and the maximum change of the $L^2(\Omega)$ -norm decay curve and of the solution profile shortly before extinction. Refining the mesh from 49×49 to 97×97 and 193×193 at fixed Δt leaves the detected t^* unchanged for all three values of λ , so the baseline mesh is already converged in space; halving and quartering Δt shifts t^* by at most 6.3×10^{-4} — about one baseline time step, which is also the quantization of the detection itself, since t^* is read off the discrete time grid — while the norm curves and pre-extinction profiles change by at most 7×10^{-4} and 2×10^{-4} , respectively. The reference extinction times are therefore reliable to three decimal places, with an uncertainty of order one baseline step in the fourth, and the monotone ordering $t^*(0.8) < t^*(0.6) < t^*(0.4)$ is preserved at every refinement level. The refinement study is part of the replication pipeline (reference stage of this experiment) and is regenerated together with all other reported numbers.

Network architecture and loss function. The trial solution takes the hard-constrained form

$$u_\theta(t, x, y) = u_0(x, y) + t x(1 - x) y(1 - y) N_\theta(t, x, y), \quad (6.26)$$

where $N_\theta : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a fully connected neural network. The parametrization (6.26) enforces the initial condition $u_\theta(0, \cdot) = u_0$ and the homogeneous Dirichlet boundary condition exactly for every θ (the polynomial factor vanishes on $\partial\Omega$), so, as in Section 6.1, the DR-PINN loss reduces to the pure

λ	grid	Δt	t^*	Δt^*	$\max \Delta \ u\ $	$\max \Delta u_{\text{pre}} $
0.4	49×49	0.0005	0.1845	+0.0000	0	0
0.4	49×49	0.00025	0.1840	-0.0005	4.5×10^{-4}	1.7×10^{-4}
0.4	49×49	0.000125	0.1839	-0.0006	6.7×10^{-4}	1.1×10^{-4}
0.4	97×97	0.0005	0.1845	+0.0000	5.1×10^{-5}	4.1×10^{-5}
0.4	97×97	0.00025	0.1840	-0.0005	5.0×10^{-4}	1.3×10^{-4}
0.4	193×193	0.00025	0.1840	-0.0005	5.1×10^{-4}	1.2×10^{-4}
0.6	49×49	0.0005	0.1645	+0.0000	0	0
0.6	49×49	0.00025	0.1643	-0.0003	4.3×10^{-4}	8.9×10^{-5}
0.6	49×49	0.000125	0.1640	-0.0005	6.4×10^{-4}	1.3×10^{-4}
0.6	97×97	0.0005	0.1645	+0.0000	5.3×10^{-5}	5.3×10^{-5}
0.6	97×97	0.00025	0.1643	-0.0003	4.8×10^{-4}	8.9×10^{-5}
0.6	193×193	0.00025	0.1643	-0.0003	5.0×10^{-4}	8.7×10^{-5}
0.8	49×49	0.0005	0.1505	+0.0000	0	0
0.8	49×49	0.00025	0.1502	-0.0003	4.1×10^{-4}	1.2×10^{-4}
0.8	49×49	0.000125	0.1501	-0.0004	6.2×10^{-4}	1.8×10^{-4}
0.8	97×97	0.0005	0.1505	+0.0000	5.6×10^{-5}	6.5×10^{-5}
0.8	97×97	0.00025	0.1502	-0.0003	4.7×10^{-4}	1.2×10^{-4}
0.8	193×193	0.00025	0.1502	-0.0003	4.8×10^{-4}	1.1×10^{-4}

Table 4: Refinement study for the reference solver: detected extinction time t^* (first grid time with an exactly zero discrete field; persistent up to T in all runs), its change Δt^* with respect to the baseline configuration (49×49 , $\Delta t = 5 \times 10^{-4}$), the maximum change of the $L^2(\Omega)$ -norm decay curve, and the maximum change of the solution profile shortly before extinction (at $0.9t^*$, interpolated to the baseline grid). At fixed Δt the detected t^* is independent of the mesh; the variation under Δt -refinement is of the order of one baseline time step and is dominated by the temporal quantization of the detection.

inclusion term: it is (5.1) with Φ as in (6.22) and $\lambda_0 = \lambda_b = 0$. Since $\Phi(s)$ is an interval of $\mathbb{R}$, the projection $\Pi_{\Phi(u_\theta(t_i, x_i))}$ needed to evaluate the distance residual is explicit, requiring no quadratic program:

$$\text{dist}^2(z, \Phi(s)) = \begin{cases} (z + \lambda)^2, & s > 0, \\ \max(|z| - \lambda, 0)^2, & s = 0, \\ (z - \lambda)^2, & s < 0, \end{cases} \quad z := \partial_t u_\theta - \Delta u_\theta.$$

With the *exact* multifunction Φ of (6.22), the interval branch of this formula is triggered only when u_θ is exactly zero in floating-point arithmetic — an event of probability zero for the smooth ansatz (6.26) — so almost every collocation point is forced onto one of the two sign branches, and the training landscape is essentially that of a discontinuous target. Our experiments retain this exact-relay configuration as a negative control, but the primary experiments train against the ε -banded relay

$$\Phi_\varepsilon(s) := \begin{cases} \{-\lambda\}, & s > \varepsilon, \\ [-\lambda, \lambda], & |s| \leq \varepsilon, \\ \{\lambda\}, & s < -\varepsilon, \end{cases} \quad \varepsilon \geq 0, \quad (6.27)$$

which applies the interval branch on the whole band $\{|s| \leq \varepsilon\}$ (so the middle case of the projection formula above is used whenever $|u_\theta| \leq \varepsilon$) and reduces to Φ for $\varepsilon = 0$.

For the banded experiments, the relaxed reaction term is Φ_ε . The map Φ_ε satisfies (P1)–(P3) with the same constant $m_0 = \lambda$: its values are nonempty, closed, convex intervals; it is upper semicontinuous (it is locally constant outside $\{|s| \leq \varepsilon\}$, and at $|s| \leq \varepsilon$ its value is already the full interval $[-\lambda, \lambda]$, which contains every nearby value); and the growth bound is unchanged. Theorem 5.3 therefore applies to the inclusion with reaction term Φ_ε .

Moreover, $\Phi(s) \subseteq \Phi_\varepsilon(s)$ for every s , hence $\text{dist}(z, \Phi_\varepsilon(s)) \leq \text{dist}(z, \Phi(s))$ pointwise, every solution of the exact relay inclusion is a solution of the ε -banded one, and the trained loss is a lower bound for the exact one. The converse fails: the ε -banded inclusion has, in general, strictly more solutions, and — unlike the exact relay, whose strong solution is unique by the monotonicity argument of Remark 6.1 — the graph of $-\Phi_\varepsilon$ is not monotone for $\varepsilon > 0$, so that argument does not carry over. Remark 6.3 below makes this loss of rigidity explicit and quantifies when it matters.

Since a smooth surrogate cannot extinguish exactly, we compare the solutions through the *band-entry time*

$$t_\varepsilon(u) := \inf \{ t \in [0, T] : \|u(t, \cdot)\|_{L^\infty(\Omega)} < \varepsilon \}, \quad (6.28)$$

evaluated on a uniform time grid of step 2.5×10^{-3} , together with its *persistent* variant (the first grid time after which the sup-norm stays below ε up to T). For the reference solution, $t_\varepsilon \rightarrow t^*$ as $\varepsilon \rightarrow 0^+$, so the band-entry time is a well-posed proxy for the extinction time that is meaningful for smooth surrogates; the sup-norm (rather than an L^2 threshold) is used because the band $\{|u| \leq \varepsilon\}$ is exactly the region where Φ_ε differs from Φ .

Remark 6.3 (Steady sliding states of the ε -banded relay). Let $\varphi \in H^2(\Omega) \cap H_0^1(\Omega)$ solve $-\Delta\varphi = 1$ in Ω , $\varphi > 0$ (the torsion function; for $\Omega = (0, 1)^2$, $\|\varphi\|_{L^\infty(\Omega)} \approx 0.0736$), and set $w_\lambda := \lambda\varphi$. Then $\partial_t w_\lambda - \Delta w_\lambda = \lambda$, so $u = w_\lambda$ is a stationary strong solution of the ε -banded inclusion $\partial_t u - \Delta u \in \Phi_\varepsilon(u)$ if and only if $\|w_\lambda\|_{L^\infty(\Omega)} = \lambda\|\varphi\|_{L^\infty(\Omega)} \leq \varepsilon$: in that case $|w_\lambda| \leq \varepsilon$ everywhere, the band branch is applied, and $\lambda \in \Phi_\varepsilon(w_\lambda) = [-\lambda, \lambda]$. Consequently, when $\lambda\|\varphi\|_{L^\infty(\Omega)} \leq \varepsilon$ the ε -banded dynamics does *not* force extinction: nonzero steady states exist inside the band, and after band entry a candidate with vanishing Φ_ε -residual may approximate any element of this sliding funnel, not necessarily $u \equiv 0$. The validation below therefore reports, for each trained network, the distance of the post-band tail both to zero and to the extremal witness w_λ , and records whether the witness is itself admissible for the given pair (λ, ε) ($\lambda\|\varphi\|_{L^\infty} \approx 0.0295, 0.0442, 0.0589$ for $\lambda = 0.4, 0.6, 0.8$). For $\varepsilon = 0$ the funnel collapses, w_λ is never admissible, and uniqueness holds as in Remark 6.1.

We also inspect branch selection by plotting the empirical cloud

$$(u_\theta(t_i, x_i), \partial_t u_\theta(t_i, x_i) - \Delta u_\theta(t_i, x_i))$$

over the three-branch graph of Φ_ε , whose middle step has width 2ε . Concentration of this cloud on the graph shows which admissible value is selected at the collocation points and provides a diagnostic specific to the set-valued formulation.

Numerical results. Figure 8 displays the $L^2(\Omega)$ -norm decay curves for the reference solver with $\lambda \in \{0, 0.4, 0.6, 0.8\}$ over $[0, T]$. The classical heat equation ($\lambda = 0$) decays exponentially and remains well above zero at $t = T = 0.3$, whereas all three relay reference solutions reach exact numerical extinction within $[0, 0.19]$, with the extinction time decreasing monotonically as λ increases.

The network is a fully connected tanh network with five hidden layers of width 128 and Glorot-normal initialization, trained for 40 000 epochs with the Adam optimizer under a hold-then-decay learning-rate schedule: the rate is held at 2×10^{-3} for the first 5 000 epochs and then decayed exponentially

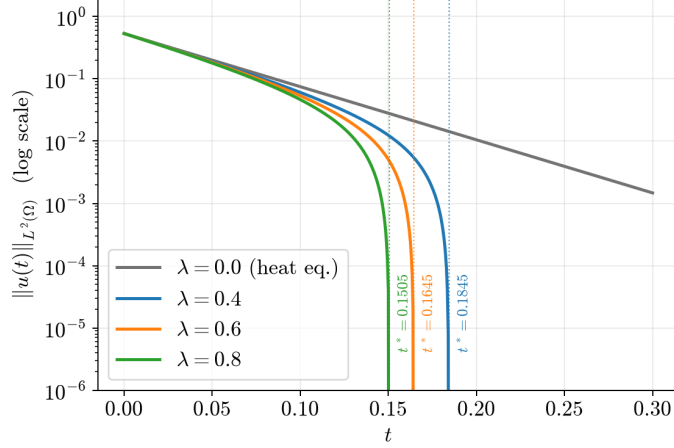


Figure 8: Reference solution: $\|u(t)\|_{L^2(\Omega)}$ on a log scale for $\lambda \in \{0, 0.4, 0.6, 0.8\}$. The vertical dotted lines mark the finite-time extinction instants t^* of the three relay cases; the $\lambda = 0$ heat-equation curve does not extinguish within $[0, T]$. Curves are clipped below 10^{-6} for readability; after extinction the reference norm is exactly zero (drop below the axis).

with rate 0.85 per 3000 steps. The same architecture and schedule are used for every configuration of the campaign.

Experimental design. The primary campaign comprises 45 independent trainings (with 5 seeds $\{11, 23, 47, 59, 71\}$ per configuration): a λ -sweep $\lambda \in \{0.4, 0.6, 0.8\}$ at fixed $\varepsilon = 0.05$; an ε -sweep $\varepsilon \in \{0, 0.01, 0.02, 0.05, 0.1\}$ at fixed $\lambda = 0.6$, with $\varepsilon = 0$ (the exact relay) serving as a negative control; and a sampler ablation (front-adaptive vs. residual-based RAR vs. uniform) at $(\lambda, \varepsilon) = (0.6, 0.05)$. After training, each network is evaluated on a dense space-time grid (time step 2.5×10^{-3}) and on a fresh cloud of 4000 collocation points: we record the band-entry times (6.28) (first and persistent), the Φ_ε distance residual, the number of cloud points violating the inclusion beyond a tolerance of 10^{-4} , the sup-norm discrepancy of the pre-band $L^2(\Omega)$ -norm curve against the reference, and the tail distances of Remark 6.3. Reference band-entry times $t_{\varepsilon, \text{ref}}$ are computed from the operator-splitting solution of the exact relay (whose refinement study, Table 4, is unchanged) on the same evaluation grid.

Figure 9 shows the training and validation curves of all 45 runs. The qualitative shape of earlier single-run experiments is reproduced uniformly: an initial plateau at $\mathcal{L} \approx 20$, an abrupt drop by roughly two orders of magnitude between epochs ≈ 5000 and ≈ 8000 , and a steady decrease thereafter.

The five $\varepsilon = 0$ runs are clearly separated from all banded runs, stagnating an order of magnitude higher (final validation mean ≈ 0.44 , seed median, vs. ≈ 0.027 at $\varepsilon = 0.05$). This separation is quantified below.

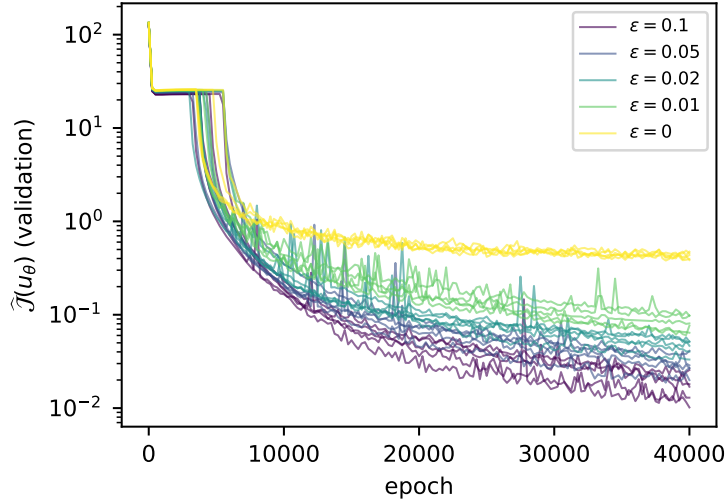


Figure 9: Training history of the inclusion loss for all 45 runs of the campaign, colored by ε (validation on a fixed batch of 4000 points). All banded runs ($\varepsilon > 0$) escape the initial plateau and continue to decrease; the five exact-relay runs ($\varepsilon = 0$, top curves) stagnate an order of magnitude higher.

Band-entry times: quantitative agreement with the reference. Table 5 reports the λ -sweep at $\varepsilon = 0.05$. For every λ and every seed the first and the persistent band crossings coincide, and the seed-median band-entry time matches the reference band-entry time *exactly* to the resolution of the evaluation grid, with seed ranges of at most one grid step (2.5×10^{-3}). In particular, the monotone ordering $t_\varepsilon(\lambda=0.8) < t_\varepsilon(0.6) < t_\varepsilon(0.4)$ of the reference is reproduced by every individual run. The ε -sweep (Table 6) shows the same agreement across $\varepsilon \in \{0.02, 0.05, 0.1\}$ — at $\varepsilon = 0.1$ all five seeds return the identical value $t_\varepsilon = 0.105$ — while at $\varepsilon = 0.01$ the agreement degrades gracefully: the median still sits on the reference value, the seed range widens to four grid steps, and for one seed the first crossing is transient (the sup-norm re-emerges above the band, and the persistent crossing occurs only at $t = 0.25$), marking the resolution limit of the smooth surrogate. At $\varepsilon = 0$ the negative control reproduces the failure mode of the exact relay documented in earlier versions of this experiment: no run achieves a persistent crossing,

two of five never cross at all, and roughly 3 900 of the 4 000 cloud points violate the inclusion beyond 10^{-4} , against ≈ 900 at $\varepsilon = 0.05$. The transition between the two regimes is sharp and occurs already at $\varepsilon = 0.02$, i.e. at a band two orders of magnitude smaller than the amplitude of the initial datum. Figure 10 visualizes the λ -sweep, and Figure 11 the residual statistics along the ε -sweep.

λ	$t_{\varepsilon, \text{ref}}$	t_ε (DR-PINN) median [range]	persistent crossings	violations (of 4 000)	tail dist. to 0	witness admissible
0.4	0.1350	0.1350 [0.1350, 0.1375]	5/5	920	1.8×10^{-3}	yes
0.6	0.1275	0.1275 [0.1250, 0.1275]	5/5	896	1.8×10^{-3}	yes
0.8	0.1200	0.1200 [0.1200, 0.1225]	5/5	867	4.0×10^{-3}	no

Table 5: λ -sweep at $\varepsilon = 0.05$ (front sampler, 5 seeds per row; violations and tail distances are seed medians). The DR-PINN band-entry time (6.28) agrees with the reference band-entry time to within one evaluation-grid step for every seed, every first crossing is persistent, and the monotone ordering in λ is preserved by every individual run — in contrast to the raw 10^{-3} threshold crossings of the exact relay reported as the $\varepsilon = 0$ control in Table 6. “Witness admissible” records whether $\lambda \|\varphi\|_{L^\infty} \leq \varepsilon$ (Remark 6.3); in all cases the trained tail is an order of magnitude closer to the zero state than to the extremal witness (cf. Table 6), so the networks select the extinct solution even where the ε -banded funnel is nontrivial.

ε	$t_{\varepsilon, \text{ref}}$	t_ε (DR-PINN) median [range]	persist. cross.	final val. mean	violations (of 4 000)	tail dist. to 0	tail dist. to w_λ
0	—	— (2/5 no crossing)	0/5	0.437	3 911	1.2×10^{-3}	2.6×10^{-2}
0.01	0.1550	0.1550 [0.1475, 0.1575]	4/5 [†]	0.081	1 301	2.4×10^{-3}	2.4×10^{-2}
0.02	0.1475	0.1475 [0.1450, 0.1475]	5/5	0.047	1 152	1.8×10^{-3}	2.3×10^{-2}
0.05	0.1275	0.1275 [0.1250, 0.1275]	5/5	0.027	896	1.8×10^{-3}	2.5×10^{-2}
0.1	0.1050	0.1050 [0.1050, 0.1050]	5/5	0.017	652	2.9×10^{-3}	2.5×10^{-2}

Table 6: ε -sweep at $\lambda = 0.6$ (front sampler, 5 seeds per row; final validation mean, violations, and tail distances are seed medians). The exact relay $\varepsilon = 0$ is the negative control and reproduces the failure mode of the discontinuous target: no persistent crossings, an order-of-magnitude larger residual, and near-total violation of the pointwise inclusion on the evaluation cloud. From $\varepsilon = 0.02$ on, band-entry times match the reference to within one grid step with persistent crossings for all seeds. [†]At $\varepsilon = 0.01$ one seed’s first crossing is transient; its persistent crossing occurs only at $t = 0.25$. The tail is consistently an order of magnitude closer to $u \equiv 0$ than to the extremal witness w_λ of Remark 6.3.

Figure 12 overlays the $L^2(\Omega)$ -norm decay curves of all five seeds against the reference for the baseline $\lambda = 0.6$ (companion figures for $\lambda = 0.4$ and 0.8 are included in the replication package). The curves coincide with the

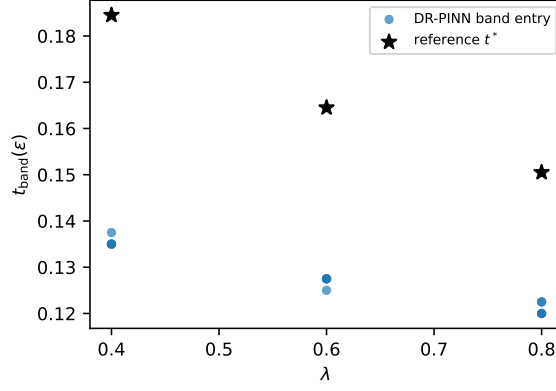


Figure 10: Band-entry times across $\lambda \in \{0.4, 0.6, 0.8\}$ at $\varepsilon = 0.05$: DR-PINN values (five seeds per λ , dots) against the reference band-entry times and the reference extinction times t^* (stars). The DR-PINN values sit on the reference band-entry values to within the evaluation-grid resolution and preserve the monotone ordering in λ run by run.

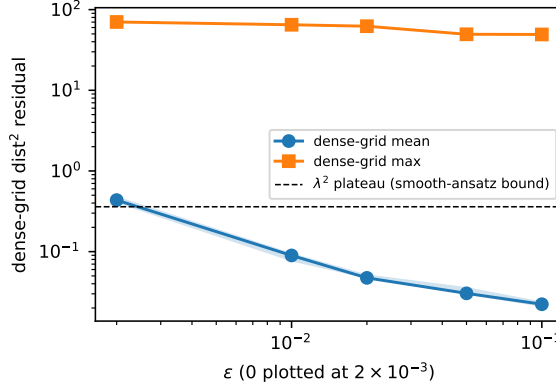


Figure 11: Dense-grid distance-residual statistics along the ε -sweep at $\lambda = 0.6$ ($\varepsilon = 0$ plotted at 2×10^{-3} on the logarithmic axis). The mean residual decreases monotonically with ε and drops below the λ^2 plateau of the smooth-ansatz bound as soon as the band opens, while the maximum residual — attained ahead of the moving interface — remains of order λ^2 times the squared operator overshoot for every ε .

reference to within $\approx 2\%$ in sup norm over the pre-band interval, for every seed and every λ of the sweep. After band entry the network norms stabilize at a small nonzero level of order 10^{-3} – 10^{-2} : a single analytic tanh-network trajectory generally cannot be nonzero before the extinction time and exactly zero on an open post-extinction interval, so a small nonzero post-band norm is expected and is not interpreted as exact extinction; the band-entry observable (6.28) is designed precisely so that this tail does not contaminate the reported times.

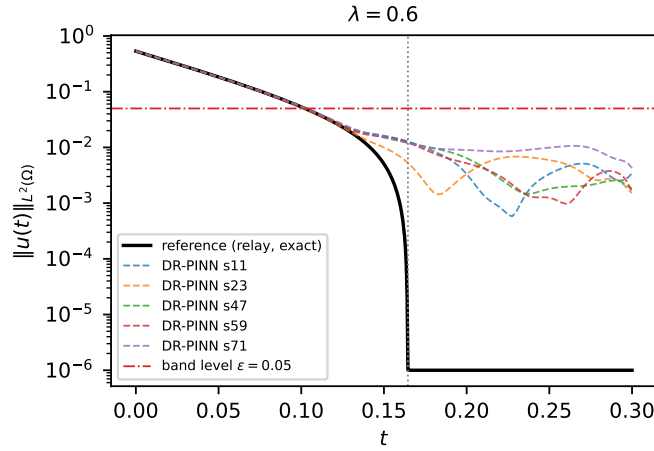


Figure 12: $L^2(\Omega)$ -norm decay for $\lambda = 0.6$, $\varepsilon = 0.05$, on a log scale: reference solution (solid) and the five DR-PINN seeds (dashed). The horizontal dash-dotted line marks the band level ε , the vertical dotted line the reference extinction time $t_{\text{ref}}^* = 0.1645$. The curves track the reference to within $\approx 2\%$ before band entry; after band entry the smooth surrogates retain a small nonzero tail, which the band-entry observable is designed to ignore.

Figure 13 shows the branch-selection diagnostic for a representative baseline run. For state values $s = u_\theta(t, x, y) > \varepsilon$, the cloud of operator values $z = \partial_t u_\theta - \Delta u_\theta$ concentrates tightly on the branch $z = -\lambda = -0.6$ of the graph of Φ_ε , confirming that the network selects the correct sign branch over most of Q ; almost no collocation points appear with $s < -\varepsilon$ (sign preservation by the relay flow, up to a small smooth undershoot near the front). Inside the band $|s| \leq \varepsilon$ the wide step $[-\lambda, \lambda]$ absorbs the operator values, and the residual there is numerically zero. The residual mass that remains — roughly 900 of the 4000 cloud points violate the inclusion beyond 10^{-4} — sits in a thin layer *ahead* of the moving interface, where $s > \varepsilon$ but the smooth output is already transitioning: there the vertical scatter of z reaches

values of order 5, far above the graph. This scatter is a direct consequence of the jump of the relay at the band edge: no smooth network output can resolve a discontinuous right-hand side pointwise.

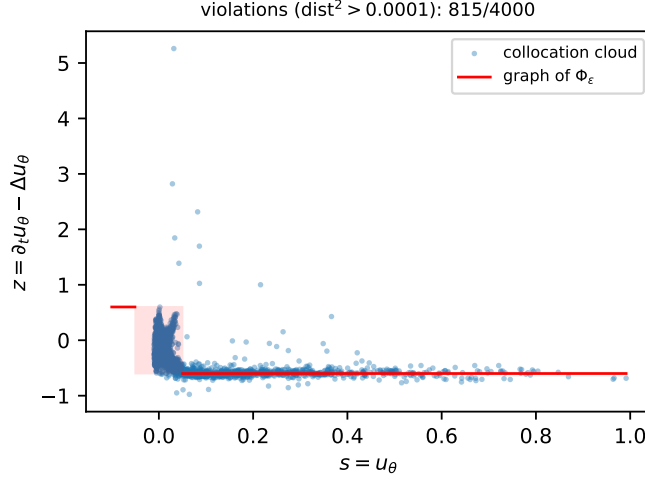


Figure 13: Branch-selection diagnostic ($\lambda = 0.6$, $\varepsilon = 0.05$): empirical cloud of $(u_\theta, \partial_t u_\theta - \Delta u_\theta)$ at 4000 post-training collocation points, overlaid on the graph of Φ_ε (red; the shaded step of width 2ε is the band). Points with $s > \varepsilon$ cluster tightly on the branch $z = -0.6$; inside the band the residual vanishes; the vertical scatter concentrates in a thin layer ahead of the moving interface, where the discontinuity of the relay limits pointwise accuracy. The inset magnifies the band region.

Sampler ablation. Table 7 compares, at the baseline $(\lambda, \varepsilon) = (0.6, 0.05)$, the front-based adaptive sampler against residual-based RAR [22] and purely uniform collocation, five seeds each. All three samplers produce identical median band-entry times (0.1275, persistent for every seed), and their residual and violation statistics overlap within seed-to-seed variability. At these settings the adaptive schemes therefore provide no measurable benefit over uniform sampling; we report this null result explicitly and retain the front-based sampler in the primary sweeps only for uniformity of protocol and as a safeguard for smaller ε , where the band is thinner relative to the domain.

Residual structure and remaining limitations. The residual reduction is not uniform over Q . On the 4000-point evaluation cloud at $(\lambda, \varepsilon) = (0.6, 0.05)$, the Φ_ε distance residual has seed-median mean ≈ 0.030 but maximum ≈ 50 : inside the band and after band entry the residual is numerically zero (post-band mean $\lesssim 10^{-6}$), so the trained tail is genuinely admissible, but ahead

sampler	t_ε median [range]	persistent crossings	final val. mean median [range]	violations median
front (adaptive)	0.1275 [0.1250, 0.1275]	5/5	0.027 [0.020, 0.041]	896
RAR	0.1275 [0.1275, 0.1275]	5/5	0.023 [0.017, 0.026]	899
uniform	0.1275 [0.1250, 0.1300]	5/5	0.026 [0.016, 0.027]	895

Table 7: Sampler ablation at $(\lambda, \varepsilon) = (0.6, 0.05)$, 5 seeds per sampler. The three schemes are statistically indistinguishable on every reported metric; the band-entry observable is insensitive to the collocation strategy at these settings.

of the moving interface — where $|u_\theta| > \varepsilon$ and the smooth surrogate must still approximate a sign branch across a thin transition layer — the residual remains concentrated and large in absolute terms. The pre-band $L^2(\Omega)$ -norm curves nevertheless track the reference to within $\approx 2\%$ in sup norm. Thus the band-entry observable (6.28) reproduces the extinction dynamics quantitatively: it is seed-stable, persistent, monotone in λ , and grid-exact against the reference for $\varepsilon \geq 0.02$. At the same time, the hypothesis $\mathcal{J}(u_n) \rightarrow 0$ of Theorem 5.3 is not realized uniformly because the continuous residual remains dominated by the pre-band interface layer. In addition, Remark 6.3 shows that at $(\lambda, \varepsilon) = (0.4, 0.05)$ and $(0.6, \{0.05, 0.1\})$ the ε -banded inclusion admits nonzero steady sliding states, so a vanishing post-band residual alone would not identify the extinct solution; the recorded tail distances show that every trained network is an order of magnitude closer to $u \equiv 0$ (median $\approx 2 \times 10^{-3}$) than to the extremal witness ($\approx 2.5 \times 10^{-2}$), i.e. the networks select the solution of the exact relay even where the relaxed problem does not force it.

7. Conclusions

We proposed Distance-Residual Physics-Informed Neural Networks (DR-PINNs) for set-valued differential laws. The usual pointwise residual is replaced by the squared distance from the differential operator to the admissible set; this distance vanishes exactly on admissible operator values and gives the classical PINN loss when the set is a singleton. On the theoretical side, we proved consistency results for both ordinary differential inclusions (Theorem 4.8) and parabolic partial differential inclusions (Theorem 5.3): any sequence of admissible candidates whose *continuous* distance-residual functional tends to zero converges, along a subsequence, to an exact solution of the target inclusion; relating the finite-collocation training loss to the continuous functional requires the additional sampling and regularity

assumptions discussed in Remark 3.2. These are conditional consistency results for the continuous functional. They do not assert the existence of a neural-network sequence with vanishing continuous residual, convergence of a gradient-based optimizer, or validity of the bridging assumptions for random or adaptive collocation, and they do not provide convergence rates. Each of these issues requires additional analysis. The proofs combine a priori estimates, compactness, weak convergence of the differential operators, and the lower semicontinuity of normal convex integrands; in view of the intrinsic non-uniqueness of solutions of inclusions, this compactness-type statement is the natural analogue of the error bounds available in the single-valued setting. On the practical side, the projection defining the loss is computable: explicitly for intervals, balls, and boxes; by a small quadratic program, combined with an offline Quickhull reduction, for polytopic sets; and by a scalar secular equation for ellipsoidal sets.

The numerical experiments cover three settings. For a linear control system with a polytopic input set, the distance residual is driven to machine-precision levels and the trained ensemble stays close to an empirically sampled reachable tube. For a planar inclusion with a rotating ellipsoidal constraint, the DR-PINN companion run reaches a collocation loss of 6.9×10^{-19} with endpoint constraints enforced exactly, while a complementary hard-admissible parametrization illustrates how the residual can instead be eliminated at the time-grid nodes by construction (with feasibility between nodes confirmed a posteriori on a dense validation grid), freeing the remaining degrees of freedom to select a particular element of the solution set. For a reaction-diffusion inclusion with relay feedback, training against an ε -banded relaxation $\Phi_\varepsilon \supseteq \Phi$ (which satisfies the same hypotheses (P1)–(P3), so the same consistency theorem applies) turns a merely qualitative agreement into a quantitative one: for $\varepsilon \geq 0.02$ the band-entry times of the trained networks are persistent, seed-stable (ranges of at most one evaluation-grid step over five seeds), monotone in the relay strength λ , and agree with the reference solver to the resolution of the evaluation grid, while the exact relay $\varepsilon = 0$ retains its failure mode and is reported as a negative control. A sampler ablation shows the adaptive collocation schemes to be statistically indistinguishable from uniform sampling at these settings. The residual nonetheless remains concentrated in a thin layer ahead of the extinction front, where a smooth surrogate must approximate a sign branch of the relay; and the relaxed inclusion admits nonzero steady sliding states whenever $\lambda \|(-\Delta)^{-1} 1\|_{L^\infty(\Omega)} \leq \varepsilon$, yet the trained tails are an order of magnitude closer to the extinct solution than to the extremal sliding state, i.e. the networks select the solution of the exact relay even where the relaxation does not force

it.

Several directions remain open. Quantitative error estimates would require additional structure, such as one-sided Lipschitz or monotonicity conditions on F or Φ , beyond the compactness-based consistency used here. State-dependent admissible sets also raise a separate backward-pass question because the projection moves with the network state. The relay experiment suggests combining the distance residual with hard-admissible parametrizations, adaptive sampling, or nonsmooth network components near discontinuities of the multifunction. Extensions to hyperbolic partial differential inclusions, sweeping processes, and differential variational inequalities are natural further directions. Relaxing convexity of $F(t, x)$ and $\Phi(u)$ is also a natural but nontrivial open problem, since it would require prox-regularity of the admissible sets in place of convexity for the differentiability of the squared distance.

Code and data availability

All numbers, tables, and figures of Section 6 are produced by a single execution of the replication pipeline referenced throughout the text. The replication package is publicly available at <https://github.com/FILL-USERNAME/dr-pinns-replication> (release v1.0-submission, commit FILL-COMMIT-HASH); a snapshot of the release is permanently archived on Zenodo under doi:FILL-ZENODO-DOI. The package contains: the scripts `run_experiment_61.py`, `run_experiment_62.py`, and `run_experiment_63.py` generating every table and figure of Sections 6.1–6.3 (including the automatically generated numerical macros `generated/results_6x.tex` compiled into this manuscript); all random seeds; the saved network checkpoints, in particular those underlying the figures and tables of Section 6.3; the exact definitions and weights of all regularizers entering the cost (6.18); and a manifest recording the hardware, operating system, and full library versions of the runs whose outputs are reported here.

Acknowledgments

This article is based upon work from COST Action InterCoML, CA24136, supported by COST (European Cooperation in Science and Technology).

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